



Model Predictive Control for Multimodal Intelligent Transportation Systems: A Cross-Domain Review of Ground, Rail, Low-Altitude, and Marine Applications



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Abstract: Intelligent transportation systems increasingly rely on connected and autonomous platforms operating across road, rail, low-altitude, and marine environments. These systems must accommodate nonlinear and time-varying dynamics, environmental uncertainty, communication limitations, safety requirements, and tightly coupled operational constraints. Model predictive control (MPC) is well suited to such conditions because it combines future-state prediction, constrained optimization, and closed-loop correction within a receding-horizon framework. This review examines the theoretical foundations, major formulations, and transportation applications of MPC across four domains: ground vehicles and traffic networks, railway systems, low-altitude unmanned aerial transportation, and marine autonomous systems. The literature was organized according to transportation mode and application, with attention given to prediction models, control objectives, operational constraints, uncertainty treatment, computational requirements, and validation methods. The review showed that ground transportation research placed greater emphasis on vehicle motion control, connected-vehicle coordination, traffic signal optimization, and network regulation. Railway applications focused mainly on train regulation, virtual coupling (VC), scheduling, energy-efficient operation, and maglev control. Low-altitude studies addressed trajectory tracking, obstacle avoidance, multi-unmanned aerial vehicle (UAV) coordination, and learning-based prediction, whereas marine studies concentrated on underactuated motion, environmental disturbances, collision avoidance, navigation rules, and surface–underwater cooperation. Across these domains, model uncertainty, online computational burden, conflicting control objectives, coupled safety constraints, and limited real-world validation remained the principal obstacles to wider deployment. The findings indicate that MPC architectures must be adapted to the dynamics, operational environment, and communication conditions of each transportation mode. This review clarifies the common and domain-specific requirements of MPC in multimodal intelligent transportation and identifies the technical issues that require further theoretical and experimental study.

Keywords: Model predictive control; Multimodal intelligent transportation systems; Connected and automated vehicles; Railway transportation; Low-altitude transportation; Autonomous marine systems; Distributed control; Constrained optimization

1 Introduction

Intelligent transportation systems are changing as autonomous control, sensing, wireless communication, artificial intelligence, and embedded computing become increasingly integrated. Transportation is no longer organized solely around isolated vehicles following predetermined rules. It increasingly involves connected and autonomous agents that interact continuously with infrastructure, other vehicles, passengers, operators, and uncertain environments. Autonomous ground vehicles must satisfy vehicle-dynamic limits, road geometry, collision-avoidance requirements, and traffic regulations simultaneously [1, 2]. Railway systems must coordinate train trajectories, operating headways, routing decisions, and energy use across individual lines and wider networks [3]. Low-altitude unmanned aerial vehicles (UAVs) operate in dynamic three-dimensional environments and must cope with nonlinear flight dynamics,

wind disturbances, payload variations, and actuator constraints [4]. Unmanned surface vehicles (USVs) and unmanned underwater vehicles (UUVs), in turn, are affected by wind, waves, currents, hydrodynamic uncertainty, restricted communication, and navigation and collision-avoidance rules [5, 6].

Transportation control methods generally include feedback control, optimization-based control, and rule-based intelligent control. Proportional–integral–derivative (PID) controllers are computationally efficient and straightforward to implement, but their performance depends strongly on parameter tuning, and operational constraints cannot usually be incorporated explicitly [7]. Sliding-mode control is resistant to disturbances and modeling errors, although chattering and the design of suitable switching surfaces remain practical concerns [8]. Fuzzy control can represent expert knowledge and nonlinear relationships without requiring a complete analytical model, but its rule base and membership functions become difficult to construct as the number of system states, control inputs, and operating conditions increase [9, 10].

These limitations are particularly evident in connected and autonomous transportation. An autonomous vehicle controller may need to handle steering-angle, steering-rate, tire-force, speed, acceleration, road-boundary, and collision-avoidance constraints at the same time [11–14]. Railway control involves speed restrictions, braking capability, train separation, station dwell time, timetable adherence, and energy consumption [15–18]. Multi-UAV operation adds three-dimensional collision avoidance, formation requirements, airspace restrictions, and communication-topology constraints. Marine transportation introduces further complications arising from environmental disturbances, underactuated vehicle dynamics, restricted waterways, and navigation regulations [19, 20]. These requirements are closely coupled: a control action intended to improve tracking accuracy or reduce travel time may also affect safety margins, passenger comfort, energy use, or transportation capacity.

Model predictive control (MPC) provides a common framework for addressing these requirements because system dynamics, control objectives, and operational constraints can be incorporated into the same finite-horizon optimization problem rather than handled through separate supervisory rules [21–24]. At each control step, MPC predicts future system behavior, determines a sequence of control actions, applies the first action, and repeats the optimization using updated measurements or state estimates. This receding-horizon procedure allows the controller to respond to disturbances and modeling errors while retaining an explicit representation of state, input, safety, and operational constraints.

The need for this capability extends beyond the motion control of an individual vehicle. At the vehicle level, MPC can coordinate trajectory tracking, speed regulation, steering, braking, collision avoidance, ride comfort, and energy consumption. At the cooperative level, it can regulate interactions among connected vehicles, virtually coupled trains, UAV formations, and autonomous vessels. At the network level, MPC can predict traffic evolution and support signal timing, perimeter control, variable speed limits, train scheduling, and coordinated vehicle operation. MPC is therefore developing from a model-based feedback controller for individual platforms into a broader optimization framework for autonomous, connected, and cooperative transportation [25–30].

Several reviews have examined MPC within individual transportation domains. Research on autonomous driving has discussed motion planning and feedback control in dynamic urban environments [31]. Reviews centered on traffic signal control have summarized MPC formulations for urban intersections and freeway networks [32]. Vehicle platooning has been examined from the perspectives of distributed model predictive control (DMPC), string stability, and cooperative control [33, 34]. In the aerial domain, Nguyen et al. [35] reviewed MPC methods for micro aerial vehicles, covering linear and nonlinear formulations, operational constraints, robustness, fault tolerance, and learning-based approaches.

These studies provide detailed accounts of particular platforms or application areas, but the literature remains divided largely along modal boundaries. Ground vehicles, railway systems, UAVs, USVs, and UUVs are commonly examined as separate control problems, even though they share several requirements, including future-state prediction, explicit constraint handling, multi-objective optimization, disturbance rejection, and real-time decision-making. Examining these applications together makes it possible to identify both common control principles and differences arising from their operating environments.

The same MPC principles lead to markedly different design problems across transportation modes. Ground vehicles require high-frequency control in dense and partly unpredictable traffic, while road-network applications must account for spatial traffic propagation and fluctuating demand. Railway systems operate on fixed infrastructure and under stringent safety rules, but train coordination, timetable regulation, and energy-efficient operation can produce large-scale or mixed-integer optimization problems. Low-altitude transportation combines fast nonlinear dynamics with three-dimensional obstacle avoidance, limited onboard resources, and multi-UAV communication. Marine systems are strongly affected by environmental uncertainty, underactuation, navigation rules, and low-bandwidth communication. These differences influence the choice of prediction model, optimization formulation, constraint representation, control architecture, and validation method.

Accordingly, this review examines the development and application of MPC across ground, rail, low-altitude, and marine transportation. Rather than treating these domains as unrelated bodies of control research, the paper considers

how MPC formulations respond to their respective dynamic characteristics, operational constraints, uncertainty sources, communication conditions, computational demands, and safety requirements. The main contributions of this review are as follows:

1. The general MPC formulation and the principal algorithmic variants relevant to intelligent transportation are summarized, with attention to their assumptions, constraint-handling capabilities, and application conditions.
2. MPC applications in ground, rail, low-altitude, and marine transportation are reviewed across individual-platform control, cooperative operation, traffic regulation, scheduling, energy management, and integrated planning and control.
3. Common features and domain-specific differences are examined in terms of system dynamics, operational constraints, uncertainty, communication, computation, safety, and practical implementation.
4. The main technical bottlenecks and engineering barriers are identified, including model mismatch, online computational burden, conflicting control objectives, coupled constraints, and the gap between simulation and physical deployment.
5. Current research directions involving learning-based MPC, distributed and hierarchical control, robust and stochastic optimization, event-triggered operation, and computationally efficient implementation are discussed in relation to the requirements of different transportation modes.

The remainder of this paper is organized as follows. Section 2 introduces the fundamental principles and major formulations of MPC, including linear, nonlinear, explicit, robust, stochastic, and data-driven approaches. Section 3 reviews MPC applications in ground transportation, covering individual-vehicle control, traffic signal optimization, road-network management, connected and automated vehicle (CAV) coordination, and other traffic scenarios. Section 4 examines MPC in railway transportation, with emphasis on train regulation and scheduling, maglev control, and emerging rail applications. Section 5 discusses trajectory tracking, obstacle avoidance, formation control, and learning-based MPC in low-altitude unmanned transportation. Section 6 reviews MPC for marine unmanned systems, including USV and UUV trajectory control, collision avoidance, integrated planning, and surface-underwater cooperation. Section 7 discusses the common challenges and future directions associated with modeling uncertainty, computational complexity, multi-objective optimization, coupled constraints, and the transition from simulation to real-world deployment. Section 8 concludes the review.

2 Review Methodology

This study adopted a structured narrative review to examine the development and application of MPC in multimodal intelligent transportation systems. Relevant literature was collected from the Web of Science Core Collection, IEEE Xplore, ScienceDirect, and China National Knowledge Infrastructure (CNKI). These databases were selected because they cover a substantial body of research in transportation, automatic control, vehicle engineering, robotics, aerospace engineering, and marine engineering.

The literature search combined MPC-related terms with terms representing the transportation domains covered in this review. The MPC-related terms included “model predictive control,” “linear MPC,” “nonlinear MPC,” “robust MPC,” “stochastic MPC,” “distributed MPC,” and “data-driven MPC.” These terms were considered together with application-related expressions such as “autonomous vehicle,” “connected vehicle,” “traffic control,” “railway,” “train control,” “maglev,” “unmanned aerial vehicle,” “UAV,” “unmanned surface vehicle,” “USV,” “autonomous underwater vehicle,” and “UUV.” The combinations were adjusted to the search functions of the respective databases.

The relevance of the collected studies was assessed from their titles, abstracts, and, where necessary, full texts. Priority was given to studies in which MPC served as a principal method for control, trajectory planning, scheduling, coordination, or operational optimization. Studies were considered relevant when they addressed at least one of the four transportation domains examined in this paper and provided sufficient information on the prediction model, control objective, constraints, control structure, or validation procedure. Publications that referred to MPC only briefly or had no direct connection with transportation systems were not included in the main analysis. Foundational studies and representative review papers were retained where they were required to explain the theoretical development of MPC.

The selected literature was organized into four application domains: ground, rail, low-altitude, and marine transportation. Within each domain, the studies were further examined according to their main application, including motion and trajectory control, traffic and network regulation, train operation and scheduling, formation and cooperative control, obstacle and collision avoidance, energy management, and integrated planning and control. The comparison focused on the prediction model, MPC formulation, control objective, constraint treatment, uncertainty handling, computational requirements, and validation method. This classification allowed the common features of MPC to be distinguished from the requirements arising from the dynamics, operating environment, communication conditions, and safety rules of each transportation mode.

3 Design of Model Predictive Control

3.1 Model Predictive Control Basic Principle

Consider a discrete-time nonlinear transportation system

$$x_{k+1} = f(x_k, u_k, \varpi_k) \quad (1)$$

where, x_k , u_k , and ϖ_k denote the system state, control input, and disturbance at the sampling instant k , respectively.

At the sampling instant k , MPC predicts the future system trajectory over a finite horizon N_p and determines the optimal control sequence by solving

$$\begin{aligned} \text{H : } \min_{\hat{u}_k} J(x_k, u_k, N_p) &= \sum_{i=1}^{N_p-1} \left(\|x_{i|k} - r_{i|k}\|_Q^2 + \|u_{i|k}\|_R^2 \right) + \|x_{N_p|k} - r_{N_p|k}\|_P^2 \\ \text{s.t. } x_{s+1|k} &= f(x_{s|k}, u_{s|k}, \bar{w}_{s|k}), s = 0, 1, \dots, N_p - 1 \\ x_{k|k} &= x_k \\ x_{s|k} &\in \chi \\ u_{s|k} &\in U \\ x_{N_p|k} &\in \chi_f \end{aligned} \quad (2)$$

where, $r_{i|k}$ denotes the reference trajectory at the i -th step of the sampling instant k , Q , R and P are positive-definite weight matrices, respectively, χ , U and χ_f are the predicted state set, the predicted control set, and the terminal state constraint set of the system, respectively.

As shown in Figure 1, at each sampling instant, a finite-horizon optimal control problem is formulated, incorporating system dynamics, input-state constraints, and a predefined cost function. Numerical iterative solvers are employed to compute the optimal sequence of control actions over the prediction window. Only the first element of the obtained control sequence is applied to the physical plant. As time advances, the prediction horizon shifts forward, and the constrained optimization problem is re-initialized and resolved iteratively at the next time step, which enables MPC to continuously compensate for disturbances and model mismatches while satisfying hard constraints throughout system evolution.

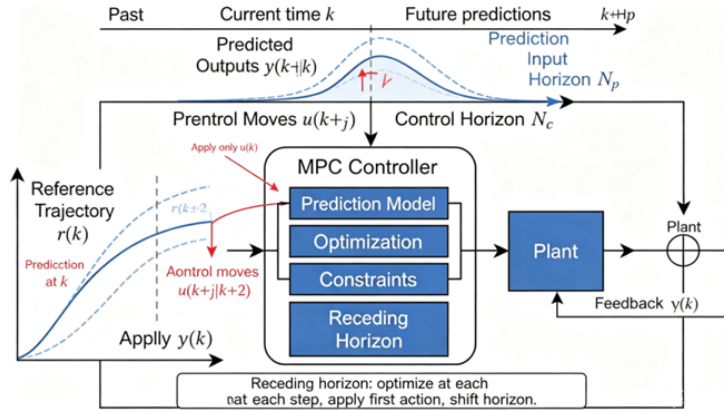


Figure 1. Schematic of the model predictive control (MPC) principle

3.2 Linear Model Predictive Control

Linear model predictive control (LMPC) is the classical formulation of MPC and has provided the theoretical and practical foundation for many subsequent MPC developments. The basic idea of MPC is to repeatedly solve a finite-horizon optimal control problem using the current state of the plant as the initial condition. For a discrete-time linear system, the prediction model can generally be written as

$$\begin{cases} x_{k+1} = Ax_k + Bu_k \\ y_k = Cx_k \end{cases} \quad (3)$$

where, x_k , u_k , and y_k denote the system state, control input, and output, respectively.

The matrices A , B , and C represent the state-transition, input, and output matrices, respectively. At each sampling instant, LMPC predicts the future system evolution over a finite prediction horizon and determines a

sequence of future control inputs by minimizing an objective function that typically penalizes tracking errors and control effort. Constraints on states, inputs, and outputs can be explicitly incorporated into the optimization problem. Only the first control action of the optimized sequence is implemented, after which the prediction horizon moves forward, and the optimization is solved again using the newly measured or estimated state. This receding-horizon mechanism provides LMPC with an implicit feedback property while retaining the advantages of optimization-based control [36, 37].

A major advantage of LMPC is its relatively low computational complexity. When the prediction model is linear, the objective function is quadratic, and the constraints are linear, the resulting optimization problem can generally be formulated as a convex quadratic program (QP), for which efficient numerical solvers are available. LMPC is therefore particularly suitable for multivariable systems with interacting inputs and outputs and for applications in which operating constraints must be handled explicitly. Unlike conventional feedback controllers, LMPC can accommodate multiple control objectives while directly enforcing actuator and state constraints. This capability supported the early development of industrial MPC, where process variables frequently operate close to safety, quality, or economic limits. Qin and Badgwell [38] documented the widespread use of MPC in chemical and petrochemical processes, refining, power generation, and other process industries. Mayne et al. [36] systematically reviewed constrained MPC and established conditions for recursive feasibility, closed-loop stability, and optimality in finite-horizon formulations.

Despite these advantages, LMPC has an inherent limitation: its prediction accuracy depends strongly on the adequacy of the linear model. A linear approximation may be satisfactory near a particular operating point, but its accuracy can deteriorate significantly when the plant exhibits strong nonlinearities, large operating ranges, saturation effects, or highly nonlinear interactions. In addition, the model parameters may vary with operating conditions, aging, environmental conditions, or unmeasured disturbances. Consequently, LMPC is most appropriate for systems whose dynamics can be reasonably represented by a linear model within the operating region of interest.

3.3 Nonlinear Model Predictive Control

Nonlinear model predictive control (NMPC) extends the MPC framework by explicitly incorporating nonlinear system dynamics into the prediction model. Instead of assuming a linear relationship between states and inputs, NMPC considers a nonlinear system of the form

$$\begin{cases} x_{k+1} = f(x_k, u_k) \\ y_k = h(x_k, u_k) \end{cases} \quad (4)$$

where, x_k , u_k , and y_k denote the system state, control input, and output at the sampling instant k , respectively, while $f(\cdot)$ and $h(\cdot)$ denote the nonlinear state-transition and output functions.

Similar to LMPC, NMPC repeatedly solves a finite-horizon optimal control problem and implements only the first control action. However, the future trajectory is generated using the nonlinear model, allowing the controller to capture nonlinear phenomena that cannot be represented adequately by a fixed linear model. The optimization problem can include nonlinear state and input constraints, nonlinear objective functions, and nonlinear terminal conditions. In this way, NMPC provides a systematic framework for combining nonlinear prediction, optimization, and constraint handling.

NMPC can provide substantially better prediction and control performance than a linearized controller, particularly when the system operates over a wide range of states and operating conditions. It has consequently been investigated extensively in automotive systems, aerospace systems, robotics, chemical processes, energy systems, and autonomous vehicles. The theoretical development of NMPC was consolidated around the late 1990s and early 2000s, with the work of Allgöwer and Zheng [39] providing an important foundation for nonlinear predictive control. Their work emphasized the relationship between NMPC, nonlinear optimal control, stability, and constrained optimization. Later, Grüne and Pannek [40] provided a systematic treatment of NMPC, including stability, feasibility, robustness, numerical optimization, and extensions without stabilizing terminal constraints.

However, in contrast to the convex quadratic programming problem encountered in standard LMPC, NMPC generally requires the repeated solution of a nonlinear programming (NLP) problem. Such problems may be nonconvex and may admit multiple local optima, making numerical convergence and real-time implementation more challenging. The computational burden becomes particularly significant when the prediction horizon is long, the system has many states and inputs, or the nonlinear model is computationally expensive to evaluate. In addition, theoretical guarantees concerning global optimality are generally difficult to obtain for nonconvex nonlinear optimization problems. Therefore, although NMPC can improve control performance through more accurate modeling, its practical applicability has historically been limited by computational requirements. The development of NMPC has consequently been closely associated with advances in numerical optimal control, automatic differentiation, real-time iteration schemes, sparse NLP solvers, and embedded optimization. These developments have substantially expanded the range of systems for which NMPC can be implemented in real time.

Recent NMPC research has also moved beyond conventional set-point tracking toward economic NMPC, distributed NMPC, stochastic NMPC, and learning-based NMPC. Nevertheless, for linear systems where the online optimization problem can be characterized analytically, an alternative approach is to move the computational burden offline.

3.4 Explicit Model Predictive Control

Explicit model predictive control (EMPC) was developed primarily to overcome the online computational burden of conventional MPC. In standard LMPC, the optimal control input must be recalculated online by solving a constrained optimization problem at every sampling instant. Explicit MPC instead solves the parametric optimization problem offline and derives the optimal control law explicitly as a function of the current state. For linear time-invariant systems with quadratic cost functions and linear constraints, the resulting optimal control law is typically piecewise affine, while the feasible state space is partitioned into a finite collection of polyhedral regions. Within each region, the control input can be calculated directly from an affine function without solving an optimization problem online.

A fundamental contribution was provided by Bemporad et al. [41], who demonstrated that the constrained finite-horizon linear quadratic control problem can be transformed into a multiparametric quadratic programming problem. The resulting state-feedback law is piecewise affine and continuous, thereby reducing online computation to region identification and evaluation of a simple affine control law. This development was particularly significant for embedded control systems, where the computational resources and sampling time available for online optimization may be severely limited. Once the explicit solution has been calculated offline, the controller can be implemented using relatively simple arithmetic operations and a region search procedure.

Therefore, EMPC has low and predictable online computational demand. It is particularly suitable for systems with fast sampling requirements, limited computational hardware, or applications in which predictable execution time is essential. Typical application areas include automotive control, aerospace systems, power electronics, robotics, and embedded systems. Furthermore, because the explicit solution reveals the structure of the optimal control law, EMPC can provide useful insight into how constraints affect the feedback policy. In contrast, the main disadvantage is the potentially large memory requirement associated with storing the polyhedral regions and their corresponding control laws. The number of regions can grow rapidly with the system dimension, prediction horizon, and number of constraints. Consequently, nominal explicit MPC becomes increasingly difficult to construct and store for high-dimensional systems. Approximate nominal explicit MPC and complexity-reduction methods have therefore been investigated to address this issue.

A further limitation is that the explicit solution is constructed for a specified prediction model, objective function, horizon, and constraint set. Substantial changes in the system dynamics or operating conditions may therefore require the controller to be recomputed offline. Moreover, nominal explicit MPC does not inherently account for model uncertainty or unmeasured external disturbances. These limitations become important when MPC is applied to transportation systems operating under changing environmental conditions and imperfect models, motivating the development of robust MPC formulations.

3.5 Robust Model Predictive Control

Robust model predictive control (RMPC) was developed to explicitly account for uncertainty and disturbances in the prediction model. In practical systems, the model used by an MPC controller is inevitably an approximation of the real plant. Model parameters may be uncertain, external disturbances may be unknown, and unmodeled dynamics may cause the actual system trajectory to deviate from the nominal prediction. If a nominal MPC controller is designed without accounting for such effects, constraints that are satisfied in the prediction may be violated in the real system. RMPC addresses this problem by designing the control law such that appropriate stability and constraint guarantees are maintained for a specified set of possible uncertainties.

An uncertain discrete-time linear system can be represented as

$$x_{k+1} = Ax_k + Bu_k + D\varpi_k \quad (5)$$

where, ϖ_k represents an unknown but bounded disturbance belonging to a predefined uncertainty set W .

Instead of optimizing only one nominal trajectory, RMPC accounts for the possible effects of admissible disturbance realizations. Several major approaches have been developed, including min-max MPC, constraint-tightening approaches, disturbance-feedback MPC, and tube-based MPC. In min-max MPC, the controller optimizes the worst-case performance over all admissible disturbances. Although this approach provides strong robustness guarantees, it can lead to a computationally demanding optimization problem. Tube-based MPC provides a more computationally attractive alternative by separating the nominal trajectory from the error dynamics and constructing a robust positively invariant “tube” around the nominal trajectory. The actual state is then guaranteed to remain within this tube despite bounded disturbances.

Mayne et al. [42] developed an important RMPC formulation for constrained linear systems subject to bounded disturbances and established robust exponential stability of a disturbance-invariant set. Earlier, Langson et al. [43] proposed a tube-based RMPC formulation in which the optimization generates a tube and associated piecewise-affine control law capable of maintaining the system trajectory within the tube under uncertainty. Importantly, the proposed approach achieved computational complexity that was manageable compared with more general min-max formulations. These contributions played an important role in establishing tube-based MPC as one of the dominant approaches to robust predictive control.

The principal advantage of RMPC is its ability to provide deterministic guarantees. If the uncertainty set is correctly specified and the underlying assumptions are satisfied, the controller can guarantee constraint satisfaction for all disturbances within the prescribed set. This property makes RMPC attractive for safety-critical applications such as autonomous systems, aerospace, automotive control, energy systems, and industrial processes. However, robustness comes at the cost of conservatism and computational complexity. If the uncertainty set is chosen excessively large, the controller may become overly conservative and sacrifice performance. Conversely, if the uncertainty description is too small, the desired robustness guarantees may no longer be valid. Furthermore, worst-case optimization may lead to unnecessarily cautious control actions because all admissible disturbances are treated as potentially occurring.

3.6 Stochastic Model Predictive Control

Stochastic model predictive control (SMPC) extends the MPC framework by explicitly modeling uncertainty using probability distributions. Instead of requiring state and input constraints to be satisfied for every possible disturbance realization, SMPC allows a controlled probability of constraint violation. A stochastic system can be represented by the model in Eq. (1), where ϖ_k is a stochastic disturbance characterized by a known or estimated probability distribution. The optimization problem can then include chance constraints such as

$$\Pr(x_k \in \mathcal{X}) \geq 1 - \varepsilon \quad (6)$$

where, ε represents an acceptable probability of constraint violation.

The central idea is therefore to explicitly balance control performance against probabilistic risk. As illustrated in Figure 2, RMPC focuses on worst-case uncertainty processing and provides hard constraint guarantees, while SMPC adopts probabilistic optimization and ensemble prediction to obtain probabilistic bounds for system outputs, which intuitively reflects the core difference between these two branches.

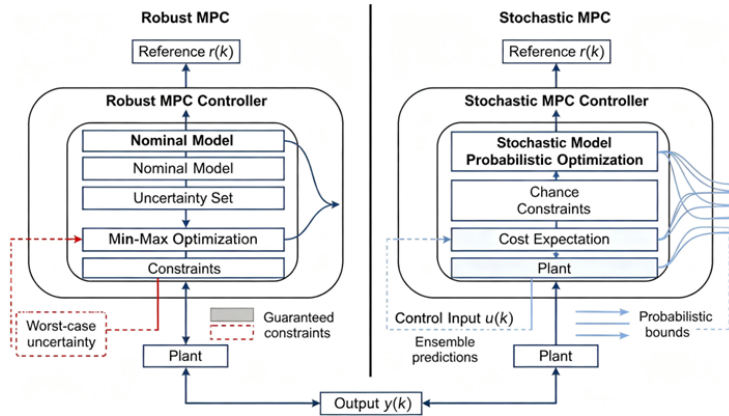


Figure 2. Robust model predictive control (RMPC) and stochastic model predictive control (MPC)

The development of SMPC was motivated partly by the conservatism that may arise when deterministic robust control is applied to disturbances for which reliable statistical information is available. For example, when a disturbance follows an approximately Gaussian distribution, designing a controller for an excessively large worst-case uncertainty set may produce unnecessarily restrictive control actions. SMPC instead uses the probability distribution of the disturbance and permits a small, explicitly specified risk of constraint violation. It may therefore achieve better expected performance and lower conservatism than RMPC, although this result depends on the accuracy of the distributional model and the selected risk level. Farina, Giulioni, and Scattolini reviewed stochastic linear MPC with chance constraints and classified existing methods according to their system dynamics, objective functions, probabilistic constraints, feasibility properties, and convergence behavior [44]. Mesbah presented a broader overview of SMPC and identified uncertainty modeling, chance-constraint reformulation, stability, recursive feasibility, and computational tractability as major research issues [45].

SMPC is particularly suitable for applications in which uncertainty can be characterized statistically and a small probability of violating selected constraints is acceptable. Relevant fields include energy systems with uncertain renewable generation, building climate control, chemical processes, autonomous systems, transportation, and economic systems. In these applications, variables such as solar irradiance, wind generation, energy demand, ambient temperature, traffic demand, and environmental disturbances may be modeled probabilistically. SMPC can incorporate these distributions into the prediction process, optimize expected performance, and impose probabilistic requirements on safety or operation.

Several limitations remain. First, the validity of the resulting guarantees depends on the accuracy of the assumed probability distribution. If the actual disturbance distribution differs substantially from the distribution used by the controller, the stated violation probability may no longer be valid. Second, chance-constrained optimization can be computationally demanding, particularly for nonlinear systems, joint chance constraints, or non-Gaussian uncertainties. Exact deterministic reformulations are available only for selected combinations of system models, probability distributions, and constraint structures; otherwise, approximations, sampling methods, or conservative bounds are required. Third, probabilistic and deterministic guarantees must be interpreted differently. A chance constraint deliberately permits a prescribed probability of violation and therefore does not generally provide the same worst-case protection as RMPC.

These difficulties have led to the development of distributionally robust MPC, scenario-based MPC, sample-based approximations, and methods for uncertainty distributions that are unknown or only partially specified. Farina and Scattolini [46], for example, developed an SMPC formulation for linear systems subject to multiplicative and potentially unbounded uncertainty. They used chance constraints and deterministic reformulations to obtain a tractable online optimization problem while accounting for the probabilistic nature of the uncertainty. Such approaches occupy an intermediate position between purely worst-case robust control and stochastic control based on a completely known probability distribution.

3.7 Data-Driven Model Predictive Control

Data-driven model predictive control (DD-MPC) represents a recent development in which information obtained directly from measured data is used to construct predictions and optimize control actions, thereby reducing the dependence on an explicitly specified first-principles model. The motivation for DD-MPC is particularly strong for complex systems whose dynamics are difficult to model accurately using conventional physical modeling approaches. Examples include highly nonlinear industrial processes, complex energy systems, autonomous vehicles, biological systems, and systems whose dynamics change over time. Instead of assuming that a sufficiently accurate parametric model is available beforehand, DD-MPC seeks to extract predictive information from historical or online input-output measurements. The overall workflow of data-driven MPC is shown in Figure 3, which consists of offline modeling using historical datasets and online rolling predictive control modules.

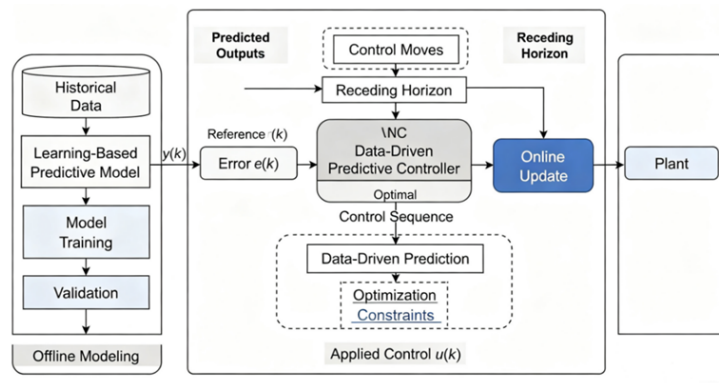


Figure 3. Data-driven model predictive control (MPC)

There are several different interpretations of data-driven MPC. One class of approaches first identifies a model from data and subsequently uses the identified model within a conventional MPC framework. These methods include subspace identification, adaptive MPC, Gaussian-process-based MPC, and machine-learning-based system identification. A more radical approach is model-free or model-minimal MPC, in which future trajectories are predicted directly from measured system trajectories without explicitly identifying a conventional state-space model. Data-Enabled Predictive Control (DeePC) is one of the most influential examples of this latter approach. Coulson et al. [47] proposed DeePC based on behavioral systems theory, where historical input–output trajectories are arranged in Hankel matrices and used to represent possible future trajectories of the system. Under appropriate conditions, the

data themselves provide a non-parametric representation of the system behavior. Importantly, the authors showed that DeePC is equivalent to classical MPC for deterministic linear time-invariant systems under suitable assumptions.

One of the major advantages of DD-MPC is its ability to reduce the modeling burden. For complex systems, developing an accurate first-principles model may require substantial domain knowledge, extensive experimentation, and considerable engineering effort. A data-driven approach can instead exploit measurements that are already available during system operation. DD-MPC can also potentially adapt to changing operating conditions if new data are continuously incorporated. This makes it attractive for systems characterized by nonlinearities, unknown dynamics, parameter variations, or complex interactions that are difficult to capture with a fixed analytical model. Data-driven methods have already been investigated in areas such as power electronics, power systems, autonomous systems, robotics, and process control. ETH Zürich, for example, has reported DeePC applications involving grid-connected power converters and quadrotor systems.

Despite its potential, DD-MPC introduces several new challenges. The quality of the controller depends strongly on the quality, richness, and representativeness of the available data. If the collected data do not sufficiently excite the relevant system dynamics, the resulting prediction may be unreliable. Measurement noise, disturbances, limited data, distribution shifts, and computational complexity can further affect performance. In addition, unlike classical model-based MPC, where stability and constraint satisfaction can often be established analytically from a known model, providing rigorous closed-loop guarantees for data-driven controllers is more difficult. Coulson et al. [48] addressed noise and uncertainty in DeePC through regularization and distributionally robust optimization, showing that regularization can be interpreted in terms of robustness against uncertainty in the measured data.

Recent research has therefore increasingly focused on establishing stability, robustness, recursive feasibility, and constraint-satisfaction guarantees for data-driven MPC. Berberich and Allgöwer [49], for example, reviewed systems-theoretic guarantees for data-driven MPC and discussed approaches based on the Fundamental Lemma of behavioral systems theory, covering settings ranging from noise-free linear systems to systems affected by noise and nonlinearities. In parallel, machine-learning-based MPC has developed rapidly, including approaches based on neural networks, Gaussian processes, reinforcement learning, and Koopman-operator representations. Recent reviews emphasize that key challenges include data scarcity, model uncertainty, generalization, computational efficiency, safety, and the establishment of closed-loop stability guarantees.

4 Model Predictive Control in Ground Transportation Systems

Ground transportation constitutes the backbone of modern intelligent transportation systems, covering autonomous individual vehicles, urban signalised intersections, large-scale road networks, and connected automated vehicle fleets. It undertakes the predominant volume of passenger and freight mobility within urban and inter-city scenarios. Distinct from three-dimensional low-altitude mobility, ground transportation operates within planar road infrastructures, yet it faces intricate coupling among vehicle dynamics, traffic participants, and network-level traffic flows. Ground systems exhibit prominent nonlinearities, time-varying traffic demand, actuator physical limits, heterogeneous driving behaviours, and multi-layer safety constraints arising from road boundaries, obstacles, and traffic rules. These complexities are amplified under congested traffic, adverse road friction conditions, and mixed traffic environments with human-driven and automated vehicles. Accordingly, advanced ground transportation calls for control methodologies capable of simultaneous trajectory execution, safety guarantees, multi-agent coordination, congestion mitigation, and robustness against time-varying environmental disturbances.

MPC stands out as a powerful control paradigm for ground transportation by virtue of its native capability to forecast future system states and explicitly enforce diverse inequality and equality constraints. At every sampling instant, MPC predicts system dynamics over a finite prediction horizon and computes an optimal sequence of control actions via constrained optimisation; only the leading control command is actuated, and the optimisation routine is reiterated with fresh measured system states. This receding-horizon mechanism permits continuous assimilation of real-time state updates and anticipative adjustment of control decisions. Compared with conventional feedback control such as PID, MPC unifies tracking performance, actuator saturation bounds, safety limitations, and control-effort penalties within one coherent optimisation formulation. A broad spectrum of MPC variants, including nonlinear, adaptive, distributed, stochastic, and data-driven MPC, have been extensively deployed across hierarchical ground-transportation scenarios ranging from single-vehicle motion control to network-wide traffic management.

The technical evolution of MPC for ground transportation reflects a stepwise enlargement of control scope. Initial investigations focus on single-vehicle control to realise accurate and stable autonomous vehicle motion. Gradually, research extends toward intersection-level traffic signal optimisation to cope with fluctuating urban traffic demand. Subsequent work targets macroscopic network traffic management for large-scale congestion alleviation, followed by cooperative control for connected and automated vehicle (CAV) groups. Most recently, data-enhanced MPC solutions are being explored to counteract model mismatch and environmental uncertainty across ground mobility applications. Such research trajectories mark a clear transition from local single-entity performance optimisation toward system-level, safe, coordinated, and adaptive ground transportation operations.

4.1 Single-Vehicle Control of Intelligent Vehicles

Trajectory tracking and path following form the foundational component for autonomous vehicle motion control. Different from classic control approaches, MPC leverages vehicle dynamics to predict future states within a prediction horizon, integrating tracking error, control input limits, and vehicle-stability constraints in optimisation to realise high tracking accuracy while maintaining desirable motion behaviour. Early research aimed to reduce lateral position and heading errors by optimising front steering angles based on kinematic or dynamic vehicle models. However, practical vehicle operation features strong nonlinearity and time-variation. Under high-speed driving and variable road friction, tyre cornering properties and yaw response change significantly, which causes model mismatch for MPC relying on fixed models and static parameters; hence, improving environmental adaptability has become a key research direction [50]. To mitigate uncertainty brought by varying road friction, some studies embed estimated friction coefficients into MPC and configure adaptive tyre slip-angle constraints, accomplishing trajectory tracking and stability regulation under a unified optimisation framework. Simulation results under high-speed steering and diverse friction conditions verify that incorporating road-friction information improves tracking accuracy and vehicle stability, implying that road-related information has evolved from external inputs into essential parts of prediction models and constraints [51]. Apart from modifying system models, another technical route focuses on adaptive tuning of objective-function weights. Li et al. [52] proposed a T-S fuzzy variable-weight MPC that adjusts weight coefficients in real-time according to lateral displacement and yaw-angle errors, enabling dynamic balance of multiple control objectives. Real-vehicle tests under double-lane-change conditions confirm its effectiveness in lowering tracking error and steering fluctuation. Building upon weight-adaptation strategies, Tang et al. [53] further presented a multi-constraint adaptive MPC with online tyre cornering-stiffness correction, which introduces vehicle stability envelope and road-environment constraints to realise integrated path-following and stability control. Tests under snow-ice pavements and abrupt friction-coefficient variation demonstrate its superior comprehensive performance. In general, trajectory-tracking MPC evolves from fixed-model schemes to online parameter adaptation, from pure tracking-error minimisation to multi-objective weight scheduling, and further to multi-constraint stability control.

As research moves beyond isolated trajectory-following to holistic vehicle motion optimisation, MPC is extended to chassis actuation systems involving steering, suspension, and direct yaw-moment control (DYC). Chassis subsystems exhibit obvious dynamic coupling across lateral, longitudinal, and vertical motions, so traditional independent control fails to reflect mutual interactions among actuators. Integrated steering-levitation MPC controllers are therefore developed for joint input optimisation. Comparative analyses show that co-optimised steering-levitation control simultaneously optimises yaw rate, body-roll angle and suspension dynamic deflection, achieving better trade-offs between handling performance and ride comfort [54]. This highlights MPC's distinctive merit: multiple subsystem performance indices can be coordinated within a single receding-horizon optimisation. Further studies target the coordination of active front steering (AFS) and DYC. For tri-axle heavy-duty vehicles, adaptive MPC is adopted to harmonise AFS and DYC, considering speed-dependent dynamic variations and obtaining improved adaptability across different velocity ranges [55]. Chassis MPC generally evolves from single-actuator control toward multi-actuator vehicle-level coordinated optimisation. Key unsolved problems include accurate modelling of complex chassis coupling, handling differences in actuator response speed and control authority, and mitigating computational burden originating from integrated optimisation.

For active-safety scenarios including autonomous emergency braking, lane keeping, obstacle avoidance and autonomous lane changing, MPC emphasises prediction of vehicle future motion and safety boundaries. By embedding collision distance, road boundary, and stability limits as optimisation constraints, proactive safety interventions can be realised. For autonomous emergency braking, existing work investigates adaptive sampling and multi-objective MPC to balance braking distance, collision risk, and ride comfort. In trajectory-tracking-cum-obstacle-avoidance tasks, yaw stability, road boundaries and obstacle constraints are merged into one optimisation formulation, promoting the integration of trajectory planning and motion control [56, 57]. Even so, the performance of safety-oriented MPC is limited by environmental perception accuracy and target prediction quality. How to handle prediction uncertainty and guarantee optimisation feasibility under multi-participant dynamic traffic represents an important research topic.

With the growing demand for high-quality driving experience, ride comfort and energy efficiency have become critical optimisation targets for single-vehicle MPC. Comfort-oriented control requires careful consideration of acceleration, jerk, yaw motion, and control-input variation, which yields prominent multi-objective characteristics. Existing research balances tracking accuracy, stability, and comfort by tuning objective-function weights, constraining input changing rates, and optimising vehicle motion states. Related autonomous-driving research has treated safety, efficiency, and comfort as joint objectives in vehicle-speed control [58]. These formulations provide useful references for defining multi-objective cost functions in MPC, although the choice of comfort indicators and their relative weights remains application-dependent. However, comfort perception varies among individuals; unified standards for quantifying comfort metrics and determining corresponding weights are still absent. Besides, energy-saving effects are heavily dependent on powertrain demand, road conditions, and driving behaviours, leading to strong

scenario-dependence for optimisation performance.

4.2 Urban Traffic Signal Optimization and Control

When applied to urban traffic signal control, MPC leverages traffic-flow prediction to forecast traffic states in a rolling horizon and dynamically optimise signal schemes according to queue length, traffic volume, and time delay. Compared with fixed-timing signal strategies, MPC can incorporate time-varying traffic demand and intersection operational states into optimisation, and its development evolves from isolated-intersection control toward multi-intersection coordination and area-wide network dynamic control. The structural differences between centralized MPC and DMPC for traffic networks are illustrated in Figure 4a and Figure 4b, respectively.

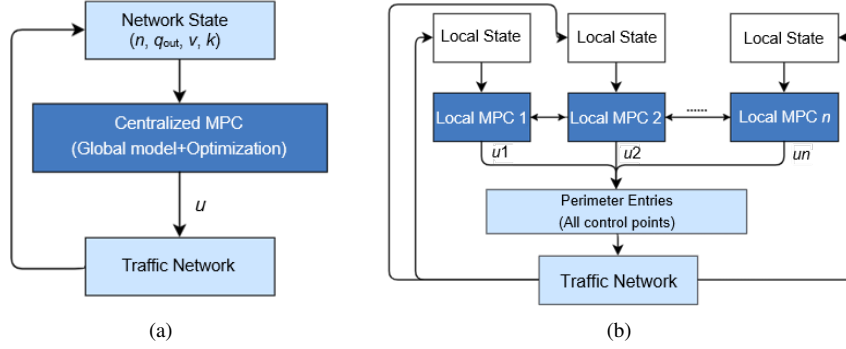


Figure 4. Model predictive control (MPC) architectures for traffic networks: (a) centralized MPC; (b) distributed model predictive control (DMPC)

At the isolated-intersection level, MPC adjusts signal timing according to real-time traffic demand to reduce vehicle queue length and average delay while enhancing intersection throughput [59]. Different traffic-flow modelling strategies have been adopted: macroscopic traffic-flow models reduce computational complexity via dynamic simplification, while high-fidelity models such as the cell-transmission model better reproduce vehicle propagation and queue evolution [60]. To handle discrete decision variables including signal phase switching and green-light duration, MPC-based signal optimisation has expanded from linear or quadratic programming to mixed-integer optimisation. Synchronous optimisation of cycle length, green-time split, and offset facilitates real-time signal adjustment responding to traffic fluctuations. A core trade-off for isolated-intersection MPC lies between traffic-flow-model accuracy and online solving efficiency. High-fidelity models describe traffic behaviour more precisely yet enlarge optimisation scale; linearised models favour real-time computation at the cost of prediction accuracy. Satisfying practical signal constraints such as minimum/maximum green time and clearance time while retaining real-time capability remains a practical bottleneck.

For multi-intersection regional signal control, strong spatiotemporal traffic coupling exists across adjacent intersections, as signal timing at one node directly influences arrival flow and queue status of downstream intersections. Centralized MPC puts all regional intersections within a single optimisation problem for global signal coordination, yet computational complexity rises sharply with network scale. To relieve such burden, DMPC has attracted increasing attention. It partitions the whole traffic network into multiple interconnected local subsystems, where each local controller conducts independent optimisation and realises global coordination via information exchange among neighbouring regions, achieving better scalability and local fault tolerance. Pham and Ahn proposed a DMPC combining perimeter boundary control and internal signal regulation, adopting the Alternating Direction Method of Multipliers (ADMM) for inter-controller coordination and lexicographic multi-objective optimisation to prioritise boundary-flow regulation before optimising internal intersection signals, which has been validated via Visualization Simulation for Traffic in Urban Road Networks-Matrix Laboratory (VISSIM-MATLAB) co-simulation [61]. Nevertheless, distributed architectures introduce new challenges, including inter-controller information exchange, communication delay, and reconciliation of local optimal solutions; there normally exists a certain loss of global optimality. Balancing computational efficiency, coordination performance, and global optimisation capability remains an open research issue.

Real-world urban traffic flow features significant stochasticity and time-variability induced by peak hours, traffic incidents, road maintenance, and large-scale events, which degrades the performance of schemes relying on fixed traffic-flow models and historical datasets. Accordingly, MPC-based signal control develops from deterministic-demand timing optimisation toward dynamic state prediction and adaptive regulation. On one hand, traffic-flow prediction modules are integrated into MPC, which continuously update arrival rate, queue length, and traffic density from detector measurements and iteratively adjust signal timing through receding-horizon optimization [62].

On the other hand, stochastic MPC and RMPC are introduced to handle uncertainties in traffic demand, turning ratio, and upstream-downstream flows by considering multiple demand scenarios; some studies further combine stochastic parameters with distributed optimisation for urban-network uncertainty management [63]. Moreover, data-driven traffic-state prediction has become an emerging trend. Traffic-prediction error sets an upper bound for MPC signal-control performance; hence, researchers start to explore the fusion between MPC and data-driven approaches such as reinforcement learning. Instead of simple substitution, hybrid frameworks are widely studied. For instance, distributed model-free adaptive predictive control transforms regional traffic dynamics into data-based representations and solves distributed optimisation with ADMM, lowering dependence on accurate mathematical traffic models. Other studies have combined macroscopic fundamental diagram (MFD)-based perimeter control with adaptive dynamic programming or explicit MPC to improve regional traffic regulation and route guidance [64, 65]. In summary, urban-signal MPC evolves from isolated-intersection local timing optimisation, to multi-intersection coordinated control, and further to dynamic prediction, uncertainty handling and data-driven adaptive control, with control architecture shifting from centralized toward distributed and hierarchical forms. Traffic-prediction error, detection noise, communication delay, and large-scale online computation still hinder real-world field deployment.

4.3 Macroscopic Road Network Traffic Management and Control

Extending beyond individual vehicles and isolated intersections, MPC for macroscopic road-network management targets aggregate network-level traffic states characterised by traffic flow, vehicle accumulation, average speed and congestion severity. The MFD, which quantifies the relationship among regional traffic demand, vehicle accumulation, and network throughput, provides vital theoretical support for network-level predictive regulation. Main research directions include urban-network perimeter control, freeway flow management with dynamic speed-limit adjustment, and multi-objective optimisation balancing traffic efficiency and environmental benefits. The operating principle of MPC-based urban road network perimeter control is depicted in Figure 5.

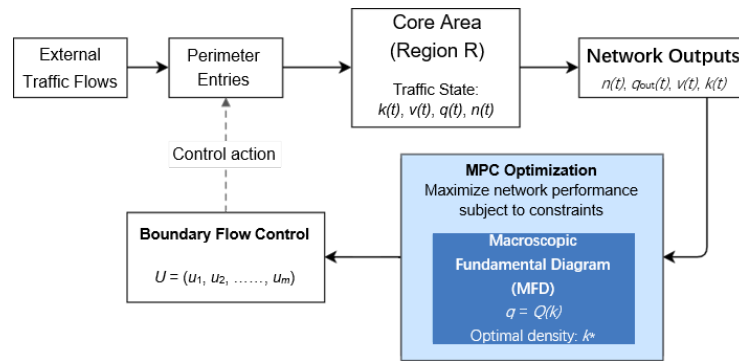


Figure 5. Model predictive control (MPC)-based perimeter control of urban road networks

Urban-network perimeter control regulates inflow traffic entering core urban zones to constrain internal vehicle accumulation and maintain network operating efficiency, typically building regional macroscopic models based on MFD and optimising boundary inflow or boundary-signal timing over receding horizons to mitigate network congestion caused by excessive traffic demand [66, 67]. Distinct from intersection-oriented signal control, perimeter control focuses on overall regional traffic conditions and implements proactive intervention before congestion forms rather than passive post-congestion response. As urban networks expand, research transitions from single-region perimeter control to multi-region coordinated schemes, treating different traffic zones as interconnected subsystems. Multiple studies further integrate intersection-level signal control and network-level perimeter control to construct multi-layer hierarchical frameworks. Under such architectures, perimeter control manages aggregate regional traffic demand, while internal signal control governs intra-network flow distribution, and the two jointly determine regional traffic evolution [61, 64]. Perimeter control therefore serves as a complementary component rather than a replacement for conventional signal control.

For freeway systems characterised by high-speed continuous flow and fast-propagating congestion waves, MPC implements active traffic regulation via variable speed limits and ramp metering [68]. Variable speed limits harmonise vehicle velocity, suppress traffic disturbances and prevent congestion in advance; ramp metering restricts vehicles merging onto main carriageways to avoid demand exceeding road capacity. Recent research evolves from separate deployment of single measures toward joint optimisation of speed limits, ramp metering and lane management. Different from conventional feedback control responding only to current traffic status, MPC explicitly considers spatiotemporal congestion propagation and adjusts control strategies proactively. For large-scale freeway networks, computational complexity arising from numerous control variables represents a major concern. In practical

application, driver compliance to speed-limit instructions, traffic-prediction error and driver-behaviour uncertainty also affect control effectiveness; incorporating driver behavioural uncertainty into prediction models constitutes a meaningful research direction.

4.4 Coordinated Control of Connected and Automated Vehicles

CAVs exchange information via vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication, acquiring traffic-state information beyond onboard sensor coverage. Against this background, MPC is extended from single-vehicle control to multi-vehicle cooperative systems, which realise joint prediction and coordinated optimisation over multiple-vehicle motion states by sharing velocity, position, acceleration, and driving-intention data. Major research fields include cooperative vehicle platooning, multi-vehicle collision avoidance, cooperative lane-changing, and intersection crossing optimisation.

Cooperative platoon control is one relatively mature CAV-coordination research branch, whose goal is to sustain stable platoon structure, improve road capacity, and maintain safe inter-vehicle gaps via information sharing. Compared with traditional car-following control, MPC can capture coupling among preceding and following vehicles during state prediction, making it well-suited for multi-vehicle car-following tasks. Early platoon-MPC studies concentrated on balancing string stability and physical safety constraints. Kianfar et al. [69] proposed a string-stable platoon MPC framework combining constraint-handling capability and linear stabilising controllers, ensuring disturbance attenuation along the vehicle string besides satisfying safety constraints. With the growth of platoon scale, centralized MPC faces severe real-time computational pressure, since prediction-horizon enlargement and more platoon members dramatically increase optimisation complexity, and control latency may even induce collision risks. To tackle this problem, deployable MPC (DMPC) and DMPC-FOA are proposed to estimate following-vehicle optimal control sequences in advance and compensate for preceding-vehicle-state prediction error, approximating ideal MPC performance with reduced computation. DMPC has thus become a vital research orientation. Liu et al. [70] designed a DMPC for flexible platoons, dividing platoons into interconnected clusters and solving local constrained optimisation via inter-vehicle communication; coupled state constraints and terminal-set conditions guarantee collision-free operation under scenarios of vehicle joining-in and leaving-out. Beyond safety and string stability, energy-economy optimisation is gradually incorporated into platoon MPC. Yu et al. [71] constructed an MPC for hybrid-electric-vehicle platoons utilising road-gradient information, integrating inter-vehicle spacing and fuel-economy terms within cost functions and achieving simultaneous improvement in safety gaps and fuel efficiency. Optimization-based collision-avoidance methods have also incorporated vehicle dynamics, safety boundaries, and interaction constraints into predictive formulations. Zhang et al. [72] developed an optimization-based approach that represented collision-avoidance requirements within a constrained vehicle-control problem.

For multi-vehicle cooperative collision avoidance, cooperative lane-changing and intersection passage scenarios, CAVs share position, velocity, and driving-intention data, which brings strong multi-vehicle dynamic coupling. Huang et al. [73] put forward a hybrid-automaton-based path-planning and cooperative-control scheme for automated-vehicle platoons, describing cooperative driving as hybrid systems with discrete manoeuvres and continuous dynamics under a distributed architecture. This approach merges artificial potential field and MPC, replacing gradient-descent optimisation with an MPC solver to handle path planning and motion control simultaneously, and builds manoeuvre-switching models for cruising and platoon-formation modes. In cooperative lane-change and merge scenarios, MPC predicts trajectories from shared information and jointly optimises velocity, acceleration and lateral motion to mitigate vehicle conflicts and unnecessary deceleration. Some studies further combine MPC with game-theoretic negotiation mechanisms to enhance cooperative decision-making capability in complex traffic. At signal-free or weakly-signalised intersections, CAVs realise cooperative crossing via V2I information exchange. Time-slot allocation and velocity optimisation reduce stopping delay and crossing conflicts, representing MPC's expansion from single-vehicle control toward traffic-system-oriented regulation. It should be noted that most existing CAV cooperative-control research assumes ideal full-information communication. Real-world conditions involve communication delay, packet loss, perception error and stochastic human-driven-vehicle behaviour. Future work should advance from idealised full-information cooperation toward robust cooperative control under incomplete-information mixed-traffic environments, together with strengthening formal safety guarantees for practical road deployment.

4.5 Applications in Other Traffic Scenarios

Low-speed autonomous vehicles and campus shuttles operate in closed or semi-closed environments such as campuses, ports and airports. Their working velocity remains relatively low, while high standards are imposed for trajectory tracking, precise stopping and riding smoothness. Low-speed scenarios impose moderate requirements for controller online computation speed, which enables adoption of high-fidelity vehicle models and complex constraint sets. MPC is utilised for path tracking, precision parking, obstacle avoidance and station stopping. Li et al. [74] proposed a composite MPC-ADRC control strategy for low-speed autonomous vehicles: MPC undertakes

feedforward optimisation, and ADRC compensates lateral tracking error caused by model simplification. Simulink-CarSim joint simulation under circular and sinusoidal trajectories validates that this composite controller reduces lateral deviation while preserving steering smoothness and ride comfort, and maintains satisfactory disturbance-rejection performance under heading-angle noise and velocity fluctuation. Low-speed automated platforms provide favourable test-beds for MPC validation. Nevertheless, when migrating toward open-road environments, improving controller adaptability against dynamic obstacles and diverse traffic participants remains a key research focus.

Comprehensive transportation hubs accommodate heterogeneous traffic participants including automobiles, pedestrians, buses and taxis, featuring multi-agent characteristics, multi-scale interaction and strong dynamic coupling. Hub-oriented control needs to jointly consider vehicle arrival, parking, passenger transfer and passenger-flow distribution; hence MPC mainly serves traffic organisation and multi-objective coordinated management. Preliminary research introduces MPC for transportation hubs and airport ground-access systems, embedding real-time traffic conditions, passenger demand, curbside congestion and parking-resource variables into predictive-control formulations [75]. However, integrated MPC frameworks synchronously coordinating vehicle flow, pedestrian flow, parking resources and transfer operations are still limited. During traffic peaks or large-scale events, MPC can adjust traffic-organisation strategies proactively according to predicted vehicle-pedestrian flows to mitigate local crowding and congestion. The major difficulty lies in distinct motion patterns and time-scales between vehicles and pedestrians: vehicle movement can be characterised by traffic-flow models, whereas pedestrian flow exhibits strong stochasticity and individual heterogeneity. Constructing unified vehicle-pedestrian coupling models and translating them into solvable MPC optimisation problems constitutes a prominent challenge. Besides, local hub traffic is tightly coupled with surrounding urban road networks, which calls for further investigation on cross-scale coordinated-control methodologies.

5 Model Predictive Control in Rail Transportation Systems

Rail transit systems feature relatively enclosed operating environments, highly identifiable dynamic behaviours, and predictable operating conditions, rendering rail transportation one of the most mature and diversified application domains for MPC. Unlike road-based transportation with highly stochastic vehicle behaviours and complex inter-vehicle interactions, railway operations are constrained by fixed physical infrastructure, standardized signalling systems, and stringent safety headway requirements. Such strongly-constrained and comparatively predictable operational characteristics align well with MPC core principles, including receding-horizon optimisation, explicit constraint handling, and predictive decision-making built upon future system information. Recent transportation research has shown that the selection of the prediction horizon strongly affects the balance between anticipatory control performance and online computational demand in MPC [76]. This trade-off is also relevant to railway applications, where control decisions must satisfy physical, signaling, scheduling, and safety constraints within limited computation and communication intervals.

Over the past decade, rapid advancements in train-to-train communication, moving-block signalling, onboard sensing and positioning technologies have promoted rail transit to evolve from the conventional manual-driving fixed-block paradigm towards intelligent operation represented by automatic train operation, virtual coupling (VC) and network-wide coordinated scheduling. MPC has become one critical enabling control technology supporting such transition. Accordingly, this section reviews representative research progress and practical deployments of MPC within rail transit from three dimensions: automated train regulation and scheduling, automatic control for maglev transportation, and other emerging rail-transit-oriented applications.

5.1 Automated Train Regulation and Scheduling

Automated train regulation and scheduling constitutes one of the most extensively investigated MPC application areas within rail transit. Existing studies span multiple operational layers, covering single-train tracking control, multi-train coordination, as well as network-level scheduling optimisation. Relevant research can be categorised into three closely interconnected branches: automatic train regulation built upon VC, network-oriented distributed cooperative scheduling for multiple trains, stations and railway lines, and multi-objective optimisation targeting punctuality, tracking safety and energy efficiency. Collectively, these directions characterise the state-of-the-art development of MPC-driven train operation and scheduling.

VC represents a flagship next-generation railway operational concept. It leverages train-to-train communication to realise cooperative multi-train operation with drastically shortened headways without mechanical coupling hardware, achieving operational effects analogous to physically coupled long train consists. Originating from vehicle platooning techniques in automated driving, VC pursues headway reduction under guaranteed operational safety to boost line capacity and transportation flexibility. Felez et al. [77] pioneered the systematic formulation of MPC-based control frameworks for virtually coupled trains. Decentralised predictive controllers were designed for leading and following trains separately, and comparative simulations against conventional moving-block signalling validated that MPC-enabled VC could substantially compress train headways while preserving safety, laying foundational support for

subsequent virtual-coupling-oriented MPC investigations.

Subsequent research further focused on robustness improvement, computational efficiency optimisation, and refined safety-constraint formulation for virtual-coupling MPC. To counteract model uncertainties and external disturbances, RMPC schemes embed uncertain variables within acceleration dynamics and braking performance to characterise deviations under realistic working conditions. Simulation outcomes demonstrate that RMPC maintains enlarged minimum safety headways under disturbed scenarios. Compared with Level 2 European Train Control System (ETCS) moving-block operation, VC achieves over 60% reduction in required following distance, striking improved trade-offs between operational safety and transport capacity [78].

Meanwhile, the sharp expansion of optimisation dimensions in centralized MPC with growing train-set scale drives the development of DMPC for high-speed virtually coupled train sets. Liu et al. [79] decomposed the original coupled optimisation task into a set of local sub-problems solvable sequentially or in parallel. Such decomposition greatly elevates online computational efficiency without sacrificing acceptable control performance and enhances real-world implementability. Following similar technical routes, Luo et al. [80] proposed distributed economic model predictive control (DEMPC) equipped with emergency-braking configurations under the space-time separation (STS) principle. By reformulating analytically intractable safety constraints as explicit linear constraints, the complicated virtually-coupled-train-set control problem is transformed into computationally tractable quadratic programming, resolving the absence of unified analytical STS-safety-constraint expressions in prior works.

Beyond safety guarantees and computational performance, MPC-based VC has been extended towards high-fidelity prediction models and event-triggered control mechanisms. Conventional analytical dynamics cannot perfectly capture preceding-train real-world motions; hence, several studies introduce Long Short-Term Memory (LSTM) networks to construct data-driven preceding-train dynamic prediction models embedded inside MPC frameworks. Corresponding simulations yield enhanced control performance versus classic dynamic models and stronger adaptability to emergency-braking and other unexpected incidents [81]. Under computation- and communication-resource-limited environments, event-triggered MPC solves optimisation problems only when predefined triggering conditions are satisfied, lowering communication overhead and computational burden while preserving precise multi-train cooperative tracking [82].

For urban-rail station-approaching scenarios where following distances dynamically shift with operational conditions and passenger comfort needs to be considered, tracking-distance models integrating traction drive systems have been established. Based on these models, NMPC tracking controllers with variable weighting coefficients and jerk constraints are designed to realise adaptive safety-headway adjustment and elevated ride comfort simultaneously [83]. More recently, learning-enhanced MPC (LMPC) and Koopman-operator-assisted NMPC have been introduced for VC control. LMPC incorporates historical operational data into optimisation workflows to learn unmodelled dynamics and cut energy consumption subject to hard constraints. Koopman-based NMPC leverages dimension-reduction and constraint-preserving lifting mappings to convert original non-convex optimal-control formulations into real-time-solvable quadratic-programming problems, offering promising pathways for deploying virtual-coupling controllers on embedded hardware. In general, MPC-for-virtual-coupling research evolves from early centralized architectures to mature methodological systems integrating robust, distributed, data-driven and event-triggered control strategies, supplying solid theoretical and technical foundations for high-density, flexible automated train operations.

Prior to virtual-coupling prosperity, abundant research had addressed cooperative tracking among multiple high-speed trains under moving-block signalling. Xun et al. [84] investigated irregular headway fluctuations induced by disturbances, which degrade line capacity and raise energy expenditure, and put forward self-triggered cooperative MPC to dynamically regulate multi-train headways. The proposed approach exhibits promising performance in boosting bottleneck-section capacity, absorbing initial delays and avoiding superfluous braking actions. From the energy-saving trajectory-planning perspective, Yan et al. [85] modelled multiple high-speed trains as communicative agents and constructed online distributed cooperative MPC frameworks. Combined with ant-colony-optimisation-based distributed cooperative optimisation algorithms, individual trains dynamically adjust trajectories by utilising available running-time margins for energy-reduction purposes. Validations using operational data collected from the Wuhan-Guangzhou high-speed railway verify its engineering practicability.

As a well-suited algorithm for distributed parallel computation, the ADMM has been widely adopted for multi-train cooperative MPC optimisation. Li et al. [86] developed an ADMM-driven distributed optimal control scheme for multiple high-speed trains, decomposing coupled global optimisation into local sub-problems solvable in parallel onboard each train, which significantly improves computational efficiency while retaining control quality. Follow-up research built symmetric ADMM-based distributed optimal operation strategies for high-speed trains jointly optimising energy efficiency and punctuality, provided theoretical convergence proof and verified effectiveness via simulation tests.

In practical multi-train cooperative operations, control objectives shift along with working conditions, e.g., transitions from normal following modes to emergency-avoidance modes. To cope with such phenomena, switching-

cost-function-based distributed model predictive control (ScDMPC) architectures are proposed [87], enabling dynamic cost-function switching responding to varying cooperative-operation requirements. Theoretical analyses prove feasibility and closed-loop stability under switching conditions, furnishing effective frameworks for cooperative control with time-varying objectives.

At the network layer, scheduling optimisation increasingly merges train control with higher-level operational variables including timetable compilation and passenger-flow distribution, generating practically oriented bi-level and hierarchical MPC architectures. Liu et al. [88] designed a bi-level metro-network MPC framework unifying timetables, passenger-flow distributions and train speed-profile optimisation. The upper layer conducts macroscopic timetable-oriented coordination, while the lower layer executes fine-grained individual-train speed-profile tracking; cross-layer information exchange realises coordinated optimisation across different operational hierarchies. Following a similar hierarchical design, Chen et al. [89] developed an MPC-based framework that combined automatic train regulation with online energy-efficient speed-trajectory generation. The upper level used a nonlinear train-regulation model incorporating dynamic passenger flows to improve timetable adherence, while the lower level generated energy-efficient speed trajectories for individual trains in a distributed manner. The two levels exchanged information on running time, onboard passenger numbers, and actual arrival times, allowing the control strategy to respond to operational disturbances.

Passenger-oriented LMPC has also been exploited for train rescheduling tasks [90]. Optimisation objectives take total passenger waiting time and train energy consumption into consideration, with train composition, departure sequence and departure moments treated as decision variables. Neural-network-MPC hybrid architectures mitigate combinatorial-explosion risks triggered by discrete decision variables, supporting network-scale scheduling optimisation balancing passenger-service quality and operational costs under realistic disturbances. Collectively, these works demonstrate that network-oriented distributed MPC is advancing beyond pure train-motion control towards integrated decision-making systems that jointly handle scheduling, train regulation and passenger-flow management.

Punctuality, tracking safety and energy consumption constitute three fundamental, inherently conflicting railway-operation performance metrics. Over-emphasis on punctuality may trigger frequent traction-braking alternations, increasing energy use and impairing passenger comfort. Excessive pursuit of energy efficiency, by contrast, will exhaust running-time margins and endanger punctuality as well as safety allowances. Hence, coordinated multi-objective optimisation for these competing indices within unified MPC formulations represents another vital research stream for automated train operation. The hierarchical control framework of virtually coupled train sets based on MPC is illustrated in Figure 6. This architecture adopts two-layer control: the upper-layer MPC performs timetable and speed-profile scheduling, whereas local leader-follower MPC modules generate control commands for individual trains. Closed-loop feedback counteracts track irregularities and aerodynamic drag disturbances, decomposing intractable multi-train coupled optimisation into distributed sub-problems.

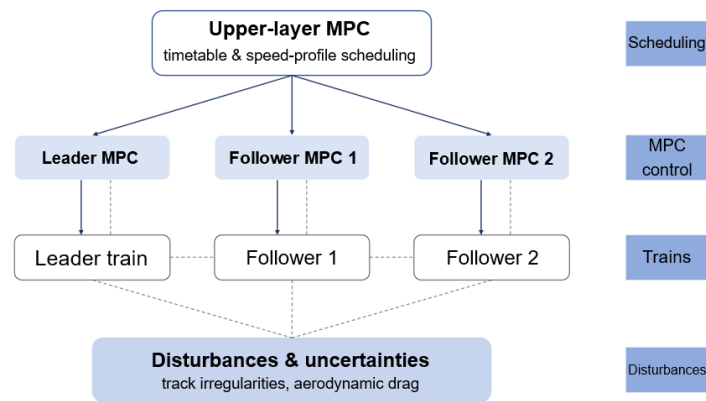


Figure 6. Hierarchical model predictive control (MPC) framework for virtually coupled train sets

For high-speed train cruise-control scenarios, Xu et al. [91] developed adaptive MPC coping with time-varying aerodynamic-drag coefficients and relative car-to-car motions. Lyapunov-stability-theory-derived adaptive parameter-update rules are deployed. Simulation results reveal superior robustness against time-varying-aerodynamic-drag-induced model mismatch compared with conventional MPC, keeping inter-car spring displacements near equilibrium positions while guaranteeing accurate speed tracking subject to safety constraints. Yang et al. [92] proposed a hybrid MPC that integrates multiple high-speed-train-operation constraints and objectives inside one optimisation formulation. Traction-braking-force allocation, operational safety, punctuality, and energy

consumption are handled simultaneously, delivering integrated automatic-operation solutions for distributed-traction-equipped high-speed trains.

Numerous studies focus on online energy-saving train speed-profile generation. Zhong et al. [93] embedded energy-saving objectives within receding-horizon optimisation to construct online high-speed-railway speed-profile generation methods. Speed trajectories are dynamically adjusted in response to real-time operational states to cut traction-energy consumption. Other researchers designed shrinking-horizon MPC trajectory-planning algorithms utilising real-time traffic information [94]. Relying on nonlinear longitudinal-dynamic-model-based future-state prediction, these algorithms optimise energy performance while respecting safety, punctuality and passenger-comfort requirements, and are particularly applicable to dynamic scenarios where ahead-of-train traffic conditions evolve during operation. With growing onboard hybrid-energy-storage-system deployment for high-speed trains, dual-layer predictive energy-management frameworks attract increasing research interest [95]. Upper-layer MPC executes long-timescale energy-allocation optimisation, and lower-layer MPC undertakes real-time torque tracking and power regulation. Adaptive observers estimate hard-to-measure internal states such as battery internal resistance and thermal degradation. Such layered structures improve energy-management flexibility and strengthen robustness against parameter uncertainties and external perturbations. At the traction-power-supply layer, Novak et al. [96] put forward hierarchical MPC for electrified-railway-traction-system energy management, jointly optimising train movement and power-supply scheduling and further expanding MPC-based multi-objective-optimisation boundaries towards railway energy management.

MPC also exhibits prominent merits for practical tracking-safety and punctuality control implementations. High-fidelity railway dynamic modelling is often challenging, while traditional single-particle or multi-particle models suffer heavy computational burdens stemming from high dimensionality. To address this gap, one investigation built MPC controllers based on Radial Basis Function-AutoRegressive with eXogenous inputs (RBF-ARX) models [97]. Massive input-output operation data logged by Automatic Train Protection (ATP) systems are utilised to construct black-box prediction models, lowering dependence on high-precision physics-driven dynamic models. Validations adopting real-world Jinan-Tai'an high-speed-railway-section operational data demonstrate enhanced train-control accuracy and passenger comfort versus classic predictive controllers.

Accurate station stopping acts as a key performance indicator for automated train operation and has been specifically researched under MPC frameworks. Liu et al. [98] designed MPC-based stopping controllers considering braking-system delays and time-varying operational conditions, taking desired speed and position as control objectives and solving quadratic-programming problems to acquire control laws. Targeting model uncertainties and limited communication resources common in high-speed-train stopping-control contexts, Liu et al. [99] developed robust self-triggered MPC schemes. Self-triggering criteria are introduced to mitigate online computation and communication loads without sacrificing stopping precision. Subsequent research incorporated coupled pneumatic-electrical-braking characteristics into system models and constructed hybrid MPC braking controllers under mixed logical dynamical frameworks [100]. These controllers jointly optimise braking-pressure-tracking accuracy and electromagnetic-valve-switching frequency, maintaining stopping accuracy and alleviating braking-system mechanical wear.

5.2 Automatic Control Applications in Maglev Transportation

Maglev transportation uses magnetic forces to provide vehicle levitation and guidance without conventional wheel-rail contact, thereby reducing mechanical wear and removing wheel-rail adhesion as a limiting factor. Its operational performance, however, depends strongly on the dynamic interaction among the vehicle, levitation system, guideway, and supporting infrastructure, particularly under track irregularities and curved-guideway conditions [101]. Nevertheless, maglev levitation systems are typically open-loop unstable and strongly nonlinear. Electromagnetic forces exhibit highly nonlinear correlations with air-gap distances, and systems are extremely vulnerable to disturbances originating from track irregularities, eddy-current effects, and load variations. Therefore, levitation-controller design permanently faces core trade-offs among stability, control performance, and explicit handling of practical engineering constraints [102].

At the single-point levitation-control level, dynamic models for electromagnetic suspension (EMS) levitation maglev single-point levitation systems have been established, accompanied by systematic stability, observability, and controllability analyses. Built upon these models, MPC levitation controllers are developed to realise stable levitation subject to electromagnet-current-saturation constraints. Linearised discrete-time models predict levitation responses, and cost functions penalise both tracking errors and control-input variations. The constrained levitation-control problem is formulated and solved as a quadratic-programming task [103]. Figure 7 depicts a classic MPC-enabled electromagnetic suspension (EMS) maglev levitation control structure. Sensor-measured airgap and velocity information is fed back to the MPC controller. Optimized control current is amplified and delivered to electromagnets, forming a closed-loop levitation regulation loop with inherent constraint-handling capability.

Following comparable technical paths, Zhang et al. [104] proposed a two-level state-feedback-assisted MPC

for magnetic-levitation systems, using measurable state variables including air gap, electromagnet acceleration, and control current. The first-layer nonlinear state feedback linearises the intrinsically unstable nonlinear levitation plant; the second-layer linear state feedback further stabilises the whole system and ameliorates dynamic responses. The two feedback layers jointly furnish stable predictive-model foundations for MPC. Simulation results verify precise air-gap regulation and favourable disturbance-rejection capabilities.

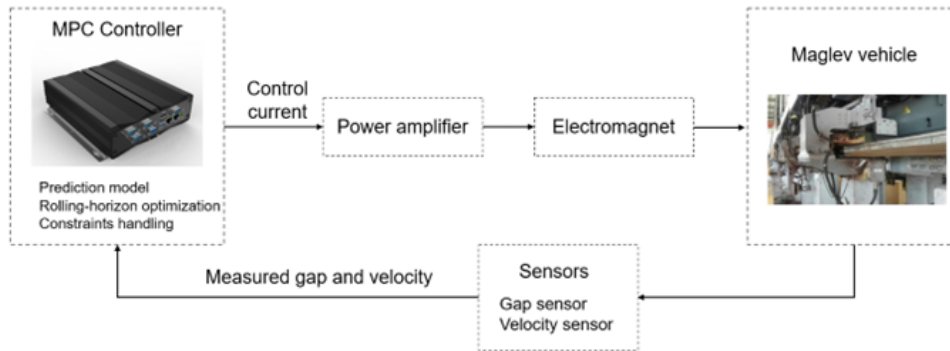


Figure 7. Typical model predictive control (MPC)-based electromagnetic suspension (EMS) levitation control structure

To strengthen robustness facing complicated external disturbances, disturbance-rejection Tube MPC strategies have been proposed [105]. State constraints for air gap, vertical velocity, and coil-control-current together with input constraints are explicitly integrated into control design. Feedback linearisation eliminates nonlinear terms existing within tracking-error dynamics. The resulting controllers realise accurate desired-air-gap tracking and preserve robust stability under disturbance scenarios.

Limited modelling accuracy of maglev levitation systems has long represented a major technical bottleneck, motivating intensive research on hybrid MPC merging data-driven techniques. For improving high-speed maglev vehicle levitation performance under complex external disturbances, LSTM-neural-network-based hybrid MPC schemes are presented [106]. LSTM networks are trained to construct nonlinear dynamic-response prediction models, upon which MPC controllers are designed targeting predefined levitation references. Moreover, PID-MPC hybrid control structures compensate for prediction-model errors induced during controller-switching procedures. Such works embody the emerging research tendency combining data-driven modelling with model-based receding-horizon optimisation.

Meanwhile, most prior investigations rely on simplified single-point levitation models and overlook modelling errors originating from real-train multi-point-levitation configurations. One study adopts multi-point levitation models and improves conventional state-constrained MPC controllers by introducing relaxation factors within cost functions, overcoming interior-point-method-derived limitations for direct state-constraint processing [107]. Utilising practical engineering parameters from the Shanghai Maglev Demonstration Line TR08 high-speed maglev vehicle, the proposed controller is evaluated under diverse operational conditions and exhibits remarkable suppression effects against severe vertical vibrations.

Beyond parameter identification and prediction-model construction, several studies explore maglev levitation-system control behaviours under faulty operational conditions. Sun et al. [108] proposed multivariate radial-basis-neural-network-based fault-tolerant control strategies for high-speed-maglev-vehicle levitation systems. Neural networks capture system nonlinearities and fault-induced influences, enhancing levitation-control stability when sensor or actuator faults occur. This progress reflects the general transition of data-driven intelligent control from normal operating states towards fault-tolerant and resilient operation.

Apart from methodological innovation, engineering implementability of MPC-based maglev levitation controllers receives growing research focus. As high-speed maglev technologies advance towards operational velocities above 600 km/h, onboard control algorithms must reliably run on resource-limited embedded hardware under strict real-time demands. Recent literature systematically investigates parameterised design and embedded deployment of MPC controllers, and processor-in-the-loop experiments assess practical algorithm performance on microcontroller hardware. Experimental outcomes prove MPC can achieve robust stabilisation for strongly nonlinear, constraint-dominated levitation systems even under high-speed conditions, supplying vital technical support for translating maglev levitation-control algorithms from numerical simulations to real-world engineering deployment [109].

5.3 Other Applications of Model Predictive Control in Rail Transit

Secondary suspension systems are installed between bogies and vehicle bodies, undertaking vibration-isolation and load-transfer responsibilities and directly determining passenger ride comfort. Nevertheless, once stiffness and damping parameters of traditional passive-suspension systems are determined, they cannot be dynamically adjusted in response to shifting operating conditions. Such inherent defects degrade passive-suspension adaptability in complicated variable-condition environments characterised by varying curve radii, track qualities, and excitation inputs [110].

To overcome these drawbacks, Orukpe [111] conducted early research applying mixed H_2/H_∞ -formulation-based MPC to railway-vehicle active suspension control. Comparative tests against classic control approaches validated MPC effectiveness for improving ride comfort. This methodology is further extended to active-vibration-control tasks for railway vehicles considering flexible-body modes. Even with car-body flexibility taken into consideration, MPC retains superior vibration-suppression performance compared with passive levitations and traditional control algorithms. These studies demonstrate that MPC can tackle multivariable control challenges within railway-vehicle active-levitation systems and leverage predictive capabilities to jointly realise vibration suppression and suspension-travel and actuator constraints, furnishing promising technical pathways for flexible-body-railway-vehicle active-vibration mitigation.

For lateral active secondary-suspension control, one prominent MPC advantage relative to conventional feedback control lies in its capacity to utilise preview information concerning upcoming track segments. Preview-enabled active-lateral-secondary-levitation control strategies are developed by building analytical dynamic models and explicitly incorporating forthcoming-track-condition impacts on car-body lateral dynamics. The corresponding MPC controller is implemented on full-scale railway-vehicle multibody-dynamic simulation models. Ride-comfort evaluation complies with EN 12299 standards, and simulation results reveal prominent continuous ride-comfort improvements, especially under small-radius-curve operating conditions [112]. Follow-up investigations evaluate controller performance under more realistic high-speed scenarios with track irregularities, and conduct systematic comparisons with mainstream active suspension control schemes including H_∞ control and sky-hook control. The preview-driven MPC solution makes better use of ahead-of-vehicle track information to realise feedforward vibration suppression and yields particularly noticeable comfort promotion on low-quality track sections [113].

With the emergence of Autonomous Rail Rapid Transit (ART) and other virtual-rail transit systems, MPC research expands towards trajectory-tracking and motion-control problems for articulated rubber-tired vehicles. Complex multi-body structures, multiple steering axles, and strong inter-unit coupling create substantial obstacles for traditional trajectory-tracking algorithms. Wang et al. [114] constructed reconfigurable Super Autonomous Rail Rapid Transit (SRT) vehicle path-tracking frameworks combining improved MPC and hierarchical control architectures. Multi-body dynamic models, generalised-force redistribution, and virtual-axle strategies are adopted to realise multi-vehicle-module steering coordination. Hardware-in-the-loop simulation tests validate satisfactory tracking accuracy, adaptability, and robustness, demonstrating MPC application potential for emerging virtual-rail-transit-vehicle trajectory control.

More recently, MPC research for virtual-track trains has further evolved towards distributed coordinated control for multi-unit vehicle systems. Sun et al. [115] established generalised dynamic-modelling frameworks for multi-unit virtual-track trains and proposed distributed MPC strategies targeting high-precision trajectory tracking. The proposed approach copes with multi-unit-vehicle-dynamics-inherent complexity while simultaneously considering trajectory-tracking accuracy and vehicle stability. Such advancements mark the transition of MPC-driven rail-vehicle control from single-car path-following to distributed predictive control for large-scale articulated virtual-rail vehicles.

6 Model Predictive Control in Low-Altitude Unmanned Transportation

Low-altitude unmanned transportation is emerging as an important component of future intelligent transportation systems, with UAVs increasingly deployed in logistics, infrastructure inspection, emergency response, surveillance, environmental monitoring, and urban air mobility. Compared with conventional ground transportation systems, low-altitude UAVs operate in a fully three-dimensional environment and possess substantially greater mobility and route flexibility. However, such advantages are accompanied by significant control challenges. UAVs are generally characterized by nonlinear and strongly coupled dynamics, underactuation, limited actuator authority, energy constraints, aerodynamic disturbances, and sensitivity to environmental uncertainties. These challenges become particularly pronounced in low-altitude environments, where buildings, trees, power lines, terrain, pedestrians, other aircraft, and other dynamic objects can significantly constrain the available flight space. Therefore, autonomous low-altitude transportation requires control strategies capable of simultaneously achieving accurate trajectory tracking, real-time obstacle avoidance, multi-UAV coordination, and robustness to uncertain and changing operating conditions.

Compared with conventional PID, feedback-linearization, and geometric control methods, MPC has the distinctive capability of simultaneously considering tracking performance, actuator saturation, state constraints,

safety requirements, and control effort within a unified optimization framework. A comprehensive survey of MPC for micro aerial vehicles further demonstrates the increasing application of linear, nonlinear, robust, distributed, and learning-based MPC to UAV control and navigation [35]. The development of MPC for low-altitude UAV transportation can therefore be understood as a progressive expansion of the control problem. Early research mainly concentrated on single-UAV trajectory tracking, where the objective was to obtain accurate and dynamically feasible motion. As UAVs began to operate in increasingly complex environments, MPC was extended to obstacle avoidance, allowing environmental and collision constraints to be incorporated into the predictive optimization. The subsequent emergence of multi-UAV applications led to formation and cooperative control, where distributed MPC was introduced to coordinate multiple vehicles under communication and collision constraints. More recently, the limitations of fixed mathematical models have motivated LMPC, in which flight data, Gaussian processes, neural networks, Koopman representations, or DeePC are used to improve the prediction model and adaptability of MPC [116, 117]. These four research directions represent an important evolution from accurate control toward safe, cooperative, and adaptive autonomous low-altitude transportation.

6.1 Unmanned Aerial Vehicle Trajectory Tracking

Trajectory tracking constitutes one of the earliest and most extensively studied applications of MPC to UAVs. The fundamental objective is to enable a UAV to follow a predefined time-varying trajectory while satisfying its dynamic, actuator, and state constraints. This problem is nontrivial because the translational and rotational dynamics of a multirotor UAV are strongly coupled, while the vehicle is underactuated. In particular, horizontal acceleration is generated indirectly through changes in attitude and thrust direction. Moreover, aerodynamic drag, wind disturbances, payload variations, and actuator saturation can lead to substantial discrepancies between simplified mathematical models and actual UAV dynamics. MPC is attractive in this context because it can predict the future evolution of the UAV and optimize control actions while explicitly considering these physical constraints.

Early research established the feasibility of implementing MPC for real-time UAV control. Bangura and Mahony developed a real-time MPC framework for quadrotors and demonstrated its capability for controlling the vehicle while accounting for its dynamic behavior [118]. Subsequently, Kamel et al. [119] developed a practical MPC framework for trajectory tracking of UAVs using ROS and experimentally demonstrated real-time implementation on a quadrotor platform. These studies were important because they established that MPC could move beyond offline optimal trajectory generation and serve as an online feedback controller for actual UAVs.

A major subsequent research direction concerned the trade-off between linear and nonlinear prediction models. Kamel et al. [120] systematically compared linear MPC and NMPC for rotary-wing micro aerial vehicles and evaluated their performance under different trajectory-tracking conditions, including aggressive maneuvers and disturbances. Their results demonstrated that linear MPC offers important computational advantages and can provide satisfactory tracking performance around nominal operating conditions, whereas NMPC can better capture the intrinsic UAV dynamics and consequently provide advantages for highly nonlinear or aggressive flight conditions. This distinction has remained fundamental in UAV MPC research: linear models are attractive for onboard implementation and high sampling rates, whereas nonlinear models become increasingly important when the vehicle operates over a wide range of attitudes, velocities, and accelerations.

The development of NMPC subsequently focused on improving computational efficiency. Kamel et al. [121] proposed a fast NMPC framework for multicopter attitude tracking on $SO(3)$, demonstrating that nonlinear predictive control could be implemented at sufficiently high rates for practical aerial-vehicle control. Wang et al. [122] further investigated efficient NMPC algorithms for quadrotor trajectory tracking and experimentally demonstrated improved computational efficiency and tracking performance. These developments were enabled by advances in numerical optimization, model discretization, sparse formulations, and efficient NLP solvers. Consequently, the computational barrier that initially limited NMPC for UAVs has gradually been reduced.

Another important direction has been the exploitation of the structural properties of quadrotor dynamics. Greeff and Schoellig developed a flatness-based MPC approach for quadrotor trajectory tracking, exploiting differential flatness to represent the system using a reduced set of variables [123]. Such formulations can substantially simplify trajectory optimization while preserving the nonlinear characteristics that are important for high-performance flight. This is particularly relevant to low-altitude transportation because UAVs may need to execute rapid changes in direction, altitude, and velocity when operating in constrained urban environments.

More recent studies have increasingly considered formal stability and tracking guarantees. Andri  n et al. [124] developed an MPC framework for quadrotors with almost-global trajectory-tracking guarantees, addressing the relationship between nonlinear UAV dynamics, predictive optimization, and closed-loop stability. This development reflects a broader transition in the field from demonstrating empirical tracking performance toward establishing theoretically rigorous properties of MPC-based UAV control.

However, nominal trajectory tracking performance can deteriorate significantly under disturbances and model mismatch. Low-altitude UAVs are affected by wind, aerodynamic drag, ground effects, payload changes, and actuator

uncertainties, all of which can reduce the accuracy of a fixed prediction model. Consequently, RMPC, disturbance estimation, adaptive modeling, and data-driven prediction have been introduced to improve tracking robustness. A particularly representative example is the data-driven MPC framework proposed by Narayanan et al. [117], which uses Koopman-based data-driven modeling to learn nonlinear quadrotor dynamics and enables MPC-based agile trajectory tracking. Their experimental results demonstrated substantial improvements in high-speed trajectory tracking, illustrating the transition from purely model-based MPC toward learning-enhanced predictive control [118–124].

6.2 Obstacle Avoidance

Obstacle avoidance represents a natural extension of trajectory-tracking MPC from purely vehicle-centric control to environment-aware predictive control. In low-altitude unmanned transportation, UAVs frequently operate in environments containing static and dynamic obstacles, including buildings, trees, power lines, vehicles, pedestrians, and other UAVs. A collision-avoidance controller must therefore not only minimize tracking error but also anticipate future collisions and modify the planned trajectory before safety boundaries are violated. MPC is particularly suitable for this problem because obstacle constraints can be incorporated directly into the predictive optimization problem.

One of the early representative studies was conducted by Dentler et al., who proposed a real-time MPC-based position controller with collision avoidance for commercial low-cost quadrotors [125]. Their approach incorporated collision avoidance directly into the MPC formulation rather than treating obstacle avoidance as an independent planning layer. Experimental implementation on a commercial quadrotor demonstrated that predictive collision avoidance could be achieved with very low computational latency [125]. This work provided an important foundation for subsequent research because it showed that trajectory tracking and obstacle avoidance could be integrated within a unified predictive controller.

The problem became considerably more challenging when multiple UAVs and uncertain vehicle states were considered. Kamel et al. [126] developed a robust NMPC framework for collision avoidance among multiple micro aerial vehicles. Their method simultaneously addressed trajectory tracking and collision avoidance while considering uncertainty in state estimation and in the predicted motion of neighboring vehicles. Experimental demonstrations with multiple MAVs showed that nonlinear predictive control could be used to achieve robust collision avoidance without relying entirely on precomputed collision-free trajectories [126]. This work represented an important transition from static and predetermined obstacle avoidance toward dynamic and uncertainty-aware predictive collision avoidance.

With the increasing deployment of UAVs in shared airspace, distributed approaches have attracted significant attention. In multi-UAV scenarios, centralized MPC requires all vehicles and obstacles to be included in a single optimization problem, causing the computational complexity to grow rapidly with the number of vehicles. Distributed MPC instead allows each UAV to optimize its own trajectory using locally available information and information received from neighboring vehicles. This architecture improves scalability and is naturally compatible with decentralized autonomous transportation systems [127].

An additional challenge arises from the uncertainty associated with dynamic obstacles. Conventional deterministic MPC generally assumes that the future position of an obstacle is known or can be represented using a deterministic safety margin. However, pedestrians, vehicles, birds, and other UAVs exhibit inherently uncertain motion. Chance-constrained MPC provides a natural solution by formulating collision avoidance probabilistically. Cao and Chi [128] proposed a chance-constrained MPC framework for dynamic UAV obstacle avoidance that explicitly incorporates uncertainty into the collision-avoidance constraints. Instead of requiring an absolutely deterministic separation, the controller constrains the probability of collision to remain below a predefined threshold. This formulation provides a more systematic mechanism for balancing safety and conservatism.

More recent work has further combined chance-constrained MPC with fast trajectory optimization. Rao et al. [129] developed a time-varying chance-constrained MPC framework for dynamic collision avoidance of quadrotors and addressed the computational challenges associated with probabilistic constraints. Such developments are particularly relevant to low-altitude transportation because the UAV may need to operate at relatively high speed while responding to dynamically changing obstacles. The use of time-varying probabilistic constraints allows the controller to account for both the temporal evolution of obstacles and uncertainty in their predicted states.

6.3 Unmanned Aerial Vehicle Formation Control

Formation control extends MPC from single-UAV autonomy to cooperative multi-UAV transportation. It is particularly relevant to future low-altitude transportation systems in which multiple UAVs may operate simultaneously for logistics, surveillance, inspection, emergency response, or coordinated transportation. Compared with single-UAV control, formation control introduces additional requirements concerning relative positions, formation geometry, inter-UAV collision avoidance, communication, and coordination. MPC is attractive because these objectives can be formulated simultaneously within a constrained optimization framework.

The theoretical foundation of distributed MPC for cooperative systems was established before its widespread application to UAVs. Richards and How [130] developed robust decentralized MPC methods for cooperative autonomous vehicles, demonstrating how distributed predictive optimization could achieve coordination while explicitly accounting for system constraints. These ideas subsequently became an important foundation for multi-UAV formation control. Early UAV formation-control studies mainly considered centralized or distributed predictive formulations in which each vehicle attempted to minimize formation errors while following a common reference trajectory. Chang and Shiau [131] investigated quadrotor formation strategies based on distributed consensus and MPC. Their work combined consensus-based coordination with MPC, illustrating an important architecture in which the consensus mechanism determines desired inter-UAV relationships while MPC determines dynamically feasible control actions.

The research subsequently moved toward integrating formation maintenance with collision avoidance and communication constraints. Rather than considering formation geometry as an isolated objective, researchers increasingly formulated it together with trajectory tracking, collision avoidance, and communication connectivity. Such an integrated formulation is important for low-altitude transportation because formation quality and safety are inherently coupled: reducing inter-UAV distance may improve formation accuracy but increase collision risk, whereas increasing separation may improve safety but reduce formation compactness and potentially weaken communication connectivity.

Distributed MPC has therefore become a particularly important research direction. Chen et al. [132] proposed a distributed MPC framework for UAV formation control under communication constraints and formulated the coordination problem as an online rolling optimization problem. Experimental results involving multiple UAVs demonstrated the ability of the proposed controller to establish and maintain the desired formation while following a reference trajectory. This work highlights the importance of explicitly considering communication limitations in practical multi-UAV systems.

Another development is the incorporation of event-triggered communication. Continuous communication among all UAVs can impose significant bandwidth and energy requirements, especially when the number of vehicles increases. Event-triggered DMPC reduces communication by transmitting updated information only when certain conditions are satisfied. Cai et al. [133] proposed a virtual-target-guidance-based distributed MPC framework for multiple UAVs, incorporating formation control and trajectory tracking while reducing unnecessary communication and computation. Such approaches are particularly attractive for low-altitude transportation because urban environments can lead to communication blockage, interference, and limited network capacity.

The literature has also investigated heterogeneous formations in which UAVs have different roles or capabilities. Zhao et al. [134] proposed a distributed coordinated-control framework based on heterogeneous roles, including leaders, coordinators, and followers, to achieve coordinated UAV swarm motion. This direction reflects the increasing complexity of practical UAV transportation systems, where different vehicles may have different payload capacities, endurance, sensing capabilities, or maneuverability.

The rapid growth of distributed MPC research has motivated several comprehensive reviews. Ouyang et al. [135] reviewed UAV swarm formation-control methods and identified distributed coordination, scalability, communication constraints, and robustness as major research issues. Peng et al. [127] subsequently reviewed distributed MPC for UAVs and vehicle platooning and categorized existing research according to trajectory optimization, formation control, collision avoidance, communication constraints, disturbances, and fault tolerance. These reviews indicate that the research emphasis has progressively shifted from simply maintaining geometric formations toward developing scalable, distributed, robust, and communication-aware cooperative control architectures.

6.4 Learning-Enhanced Unmanned Aerial Vehicle Model Predictive Control

Although conventional MPC provides an effective mechanism for trajectory tracking, obstacle avoidance, and formation control, its performance fundamentally depends on the accuracy of the prediction model. This limitation is particularly significant for low-altitude UAVs because their dynamics are affected by aerodynamic drag, wind, ground effects, payload variations, battery depletion, actuator nonlinearities, and other phenomena that are difficult to capture using a fixed analytical model. LMPC has therefore emerged as a promising research direction that combines the constraint-handling and optimization capabilities of MPC with the adaptability of machine learning and data-driven modeling.

One of the early representative studies was conducted by Bouffard et al. [136], who investigated learning-based MPC for quadrotor control and demonstrated onboard implementation and experimental validation. The fundamental concept was to retain MPC as the control framework while using learning to improve the model or compensate for discrepancies between the nominal model and the actual UAV dynamics. This architecture is particularly attractive for UAVs because it avoids the disadvantages of completely replacing model-based control with a black-box learning controller.

A major subsequent development was the introduction of data-driven predictive control. Coulson et al. proposed

DeePC, which directly uses historical input-output trajectories to predict future system behavior rather than requiring an explicitly identified parametric model [47]. Elokda et al. [137] subsequently demonstrated DeePC on physical nano-quadrotors and showed that measured trajectory data could be used to construct a predictive controller for UAV position control. This research is significant because it provides an alternative to conventional system identification and demonstrates the possibility of constructing predictive control directly from experimental flight data.

Another influential direction is learning residual or unmodeled dynamics. Narayanan et al. [117] developed a Data-Driven MPC framework for quadrotors in which Gaussian Processes were used to learn aerodynamic effects that were not adequately represented by a conventional model. The learned model was incorporated into MPC and evaluated experimentally during aggressive and high-speed flight. The authors reported substantial reductions in trajectory-tracking error compared with a conventional linear-drag model. This study is particularly relevant to low-altitude transportation because aerodynamic effects become increasingly important when UAVs operate at high speed or under aggressive maneuvers.

Neural networks have also been incorporated into UAV MPC. Jiang et al. [138] developed a neural-network-based MPC framework for quadrotor UAVs in which flight data were used to identify UAV dynamics and the learned model was subsequently integrated into MPC. Experimental and simulation results demonstrated improved trajectory-tracking performance relative to conventional control methods. The main advantage of neural-network models is their ability to approximate highly nonlinear dynamics; however, their integration into MPC can substantially increase the computational complexity of the online optimization.

A further development is the use of Koopman operator theory to construct data-driven linear representations of nonlinear UAV dynamics. Narayanan et al. [117] proposed an SE(3) Koopman-MPC framework for quadrotor UAVs, using data-driven lifting to obtain a linear predictive representation of nonlinear UAV dynamics. The resulting model was integrated into MPC and demonstrated high-rate control of agile quadrotor trajectories. This approach is particularly promising because it seeks to combine the expressive capability of data-driven modeling with the computational efficiency and mature theoretical properties of linear MPC.

Learning has also increasingly been combined with reinforcement learning. Rather than replacing MPC completely, reinforcement learning can be used to learn unknown dynamics, tune MPC parameters, generate reference trajectories, or improve higher-level decision-making. Zanon and Gros [139] proposed a safe reinforcement learning framework that combines reinforcement learning with RMPC, using RMPC to provide safety guarantees while retaining the learning capability of reinforcement learning. More broadly, Hewing et al. [116] reviewed learning-based MPC and emphasized the potential of combining machine learning with predictive control while retaining safety and constraint-handling mechanisms. For UAVs, this hybrid approach is particularly attractive because pure reinforcement learning may provide strong adaptability but lacks the explicit constraint-handling and safety properties inherent to MPC.

7 Model Predictive Control in Marine Unmanned Systems

Marine unmanned systems have become important components of intelligent ocean exploration, maritime transportation, environmental monitoring, and offshore operations. Within the scope of this review, these systems are considered in two principal categories: USVs and UUVs, with autonomous underwater vehicles (AUVs) representing an important subclass of UUVs. USVs operate on the water surface and are widely used for hydrographic surveying, environmental monitoring, maritime surveillance, search and rescue, autonomous transport, and offshore inspection. UUVs operate below the water surface and are commonly deployed for seabed mapping, underwater infrastructure inspection, oceanographic observation, and subsea intervention. The complementary capabilities of surface and underwater platforms have also led to growing interest in heterogeneous marine robotic systems, in which USVs and UUVs cooperate in missions that cannot be performed efficiently by a single platform [140, 141].

Compared with terrestrial autonomous vehicles, marine unmanned systems operate in highly uncertain and dynamically changing environments. Wind, waves, ocean currents, hydrodynamic parameter variations, actuator imperfections, and poorly known external disturbances can significantly affect vehicle motion. Furthermore, many marine vehicles are inherently underactuated, such that the number of independently controllable degrees of freedom is smaller than the number of motion variables. Consequently, their dynamics are generally characterized by strong nonlinearities, coupling effects, inertia, and nonholonomic constraints. For UUVs, these difficulties are further compounded by three-dimensional motion, limited underwater localization accuracy, and the low-bandwidth and high-latency characteristics of underwater acoustic communication [140, 142]. USVs, meanwhile, must additionally operate in environments populated by other vessels, static obstacles, restricted waterways, and dynamically changing maritime traffic.

These characteristics make MPC well suited to the control of marine unmanned systems [140, 143]. Over the past two decades, MPC research in this field has progressed from trajectory tracking and motion regulation for individual vehicles to more complex navigation and cooperative-control problems. Subsequent studies incorporated nonlinear dynamics, actuator saturation, environmental disturbances, and robustness requirements. MPC was later applied to

collision avoidance and motion planning, where predicted trajectories could be used to anticipate encounters with other vessels and obstacles before control actions were implemented. More recent research has examined integrated planning and control, distributed MPC, heterogeneous marine-vehicle cooperation, event-triggered control and optimization, and data-driven predictive control [140, 144]. Figure 8 presents a conceptual MPC architecture synthesized from the studies reviewed in this section. It brings together trajectory tracking, COLREGs-compliant collision avoidance, compensation for wind, wave, and current disturbances, and communication-constrained coordination between USVs and UUVs. The architecture covers both individual-vehicle control and cooperative mission planning, while intermittent and low-bandwidth underwater acoustic links are considered in USV–UUV coordination.

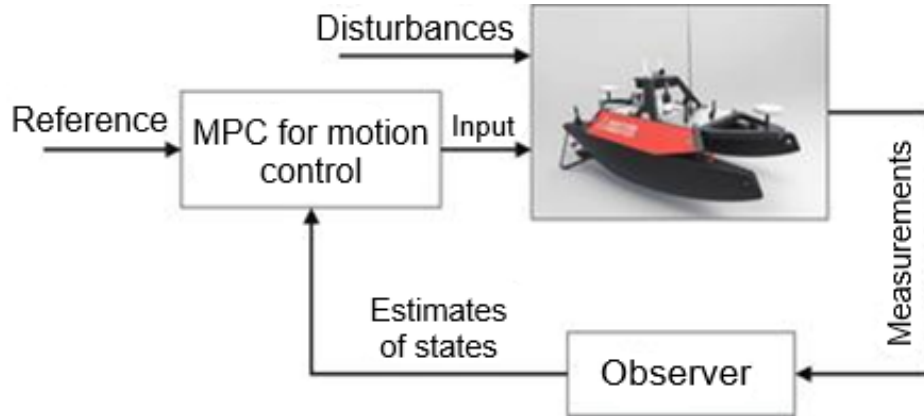


Figure 8. Model predictive control (MPC) architecture for heterogeneous marine unmanned systems

7.1 Unmanned Surface Vehicle Trajectory Tracking

One of the early studies on MPC-based vessel trajectory tracking was conducted by Zheng et al. [145], who compared nonlinear MPC with an MPC formulation based on successive linearization. A nonlinear three-degree-of-freedom vessel model was used to describe the vessel dynamics, while successive linearization was employed to formulate a computationally tractable optimization problem at each sampling instant. The reported results showed that both formulations achieved satisfactory trajectory-tracking performance while explicitly accounting for system constraints. The comparison also illustrated the practical trade-off between retaining nonlinear model dynamics and reducing the computational burden of online optimization, which remains an important consideration in real-time MPC implementation for autonomous vessels [145].

As research progressed, the limitations of nominal MPC models became increasingly evident. Marine vehicles are subject to substantial external disturbances arising from wind, waves, and ocean currents, while hydrodynamic parameters may vary significantly with operating conditions. Consequently, subsequent research increasingly incorporated disturbance observers, adaptive mechanisms, and RMPC formulations. These approaches attempt to compensate for the mismatch between the nominal prediction model and the actual vehicle dynamics, thereby improving tracking robustness in realistic marine environments [140, 144].

More recent research has introduced virtual-reference and disturbance-observer-based MPC frameworks. A virtual-USV-guided predictive control strategy, for example, was proposed to improve the transient tracking behavior of USVs by generating a virtual reference trajectory between the initial state and the desired trajectory [146]. This approach can reduce large initial tracking errors and mitigate oscillatory behavior. Such studies represent an important transition from conventional trajectory-error minimization toward predictive reference generation and transient-performance optimization.

Autonomous berthing has provided another important application scenario for USV trajectory-tracking MPC. Berthing requires high-precision low-speed maneuvering in confined waters and is particularly sensitive to wind and current disturbances. Recent research has applied NMPC to four-degree-of-freedom ship berthing and combined predictive control with state estimation to improve tracking accuracy under nonlinear environmental effects [147]. Other studies have investigated model-based trajectory planning and MPC for autonomous unberthing in constrained harbors, explicitly considering the maneuverability and dynamic limitations of underactuated USVs [148].

The research frontier has subsequently expanded toward fault-tolerant, event-triggered, and data-driven MPC. Fault-tolerant MPC has recently been developed for USV trajectory tracking and autonomous berthing while considering external disturbances, thruster failures, nonlinear dynamics, actuator saturation, and obstacle constraints simultaneously [149]. Event-triggered MPC has also been proposed for USVs operating in constrained waterways, with the objective of reducing unnecessary controller updates while maintaining tracking performance and recursive

feasibility [150]. In parallel, data-driven MPC has been investigated to overcome the limitations of fixed physics-based models when USVs operate under time-varying environmental disturbances [151].

7.2 Collision Avoidance

Johansen et al. [152] proposed one of the representative early MPC-based approaches to autonomous ship collision avoidance using scenario-based model predictive control (SB-MPC). Instead of relying on a single deterministic prediction of surrounding vessels, the method evaluated multiple possible future scenarios and optimized collision-avoidance maneuvers over a finite horizon. The approach was able to incorporate uncertainty into the predicted motion of other vessels and evaluate candidate maneuvers according to collision risk and navigational requirements [152]. This work established scenario-based MPC as an important framework for maritime collision avoidance under uncertainty.

Subsequent studies increasingly emphasized the explicit incorporation of maritime navigation regulations. MPC-based collision-avoidance frameworks were developed using ship maneuvering models together with collision-risk assessment and COLREGs-based decision-making [153]. These studies demonstrated an important advantage of MPC over purely geometric collision-avoidance methods: navigational rules, maneuverability constraints, safety distances, and control limitations can be represented directly within the optimization framework.

The next major development concerned uncertainty-aware collision avoidance. The future trajectory of a target vessel cannot generally be predicted precisely because its control intentions are unknown and its motion is affected by environmental disturbances. Deterministic prediction can therefore result either in unsafe decisions or excessively conservative avoidance maneuvers. Scenario-based approaches address this issue by considering multiple possible future trajectories [152].

The concept of informed or collaborative scenario-based MPC further advanced this research. By incorporating information regarding the intended trajectory or maneuver of other vessels, the controller can predict future interactions more accurately than methods relying solely on instantaneous position and velocity [154]. The resulting framework can coordinate collision-avoidance decisions across head-on, crossing, overtaking, and multi-vessel encounter scenarios while incorporating COLREGs requirements.

Research subsequently shifted toward real-time multi-ship collision avoidance. Zhang et al. [155] developed a multi-ship decision-making framework that combined MPC with velocity-obstacle concepts, ship maneuverability, COLREGs, and uncertainty in ship motion. This work represents an important transition in the field, because collision avoidance was formulated not simply as a geometric separation problem but as a multi-objective decision-making problem involving safety, maneuverability, uncertainty, and regulatory compliance.

More recent research has investigated intention-aware and distributed collision avoidance. Intention information has been incorporated into MPC together with quaternion-based ship-domain representations to improve collision-avoidance decisions in complex encounters [156]. Distributed MPC has also been proposed for autonomous ships operating on inland waterways, enabling multiple vessels to coordinate their collision-avoidance decisions while maintaining compliance with traffic regulations [157].

7.3 Unmanned Underwater Vehicle Trajectory Tracking

Zhang et al. [158] developed an MPC-based three-dimensional trajectory-tracking method for AUVs operating in complex ocean environments. The controller formulated trajectory tracking as a constrained optimization problem and explicitly incorporated state and input limitations. By optimizing control increments, smoother control actions could be obtained while the receding-horizon mechanism provided additional compensation against model mismatch [158]. This work demonstrated the suitability of MPC for the intrinsically three-dimensional and constrained motion of UUVs.

Subsequent research focused increasingly on external disturbances and system stability. Yan et al. [159] proposed a double-closed-loop MPC architecture for AUV trajectory tracking in the presence of external disturbances. The outer loop addressed position tracking while the inner loop regulated velocity, and system constraints were explicitly incorporated into the predictive controller. A Lyapunov-based stability analysis was also developed. The study demonstrated that a hierarchical MPC architecture could improve both tracking robustness and computational tractability [159].

The next stage of development involved RMPC. Underwater hydrodynamic parameters, including added mass and damping coefficients, are difficult to identify accurately over the entire operating envelope. In addition, ocean currents can vary significantly with depth and location. Consequently, nominal MPC may lose performance when the prediction model deviates significantly from the actual vehicle dynamics. Robust tube-MPC formulations were subsequently developed to account explicitly for model uncertainty and external disturbances [160]. Such approaches constrain the predicted system within a robust invariant tube around the nominal trajectory, thereby providing improved robustness and theoretical feasibility guarantees.

Adaptive and event-triggered approaches have also attracted increasing attention. Adaptive integral event-triggered NMPC has been investigated for AUV homing and trajectory tracking, with the objective of reducing unnecessary controller updates and communication while maintaining tracking performance [161]. Such methods are particularly attractive for underwater vehicles because computational resources and communication bandwidth are more constrained than in many surface applications.

Linear parameter-varying (LPV)-based MPC represents another important development. Instead of using a fully nonlinear model, linear-parameter-varying representations can approximate nonlinear UUV dynamics through scheduling variables while retaining the computational structure of linear MPC. LPV-MPC has therefore been investigated for AUV path planning and control [162]. This approach provides an intermediate solution between computationally inexpensive linear MPC and computationally intensive NMPC.

More recent work has explored robust quasi-linear parameter-varying MPC for UUV trajectory tracking. Such methods incorporate external disturbances and input constraints while using quasi-linear parameter-varying (qLPV) models to capture nonlinear dynamic characteristics with lower computational complexity than full NMPC [163].

7.4 Integrated Planning and Control

Traditional autonomous-navigation architectures often separate motion planning and feedback control. A planner generates a collision-free geometric path, after which a lower-level controller attempts to track the generated reference. Although this architecture is modular and computationally convenient, the planned path may not necessarily be dynamically feasible for an underactuated marine vehicle. For example, a geometrically feasible path may contain curvature or heading changes that exceed the vessel's maneuverability. MPC provides an effective mechanism for bridging this gap because trajectory generation and control actions can be optimized simultaneously.

Wei and Shi provided a comprehensive review of MPC-based motion planning and control for autonomous marine vehicles and emphasized that MPC can integrate guidance, planning, and control within a unified optimization framework [140]. This perspective is particularly important for marine vehicles because their large inertia, underactuation, and environmental disturbances make purely geometric planning insufficient for autonomous navigation.

Autonomous berthing is one representative application of integrated planning and control. In conventional systems, a path planner first generates a berthing trajectory and a controller subsequently tracks it. However, berthing requires simultaneous consideration of vessel position, heading, residual velocity, actuator saturation, wind and current disturbances, and final docking configuration. Model-based planning and MPC-based control have therefore been combined to generate trajectories that are both geometrically feasible and dynamically executable [148].

Event-triggered trajectory replanning represents another important development. In constrained harbor environments, a previously generated trajectory may become infeasible when obstacles or environmental conditions change. An event-triggered motion-planning and NMPC framework has therefore been developed in which an initial trajectory is generated offline and online replanning is activated only when specific events occur [164]. This strategy reduces unnecessary computation while preserving the adaptability of the navigation system.

More recent work has moved toward constraint-embedded MPC planning, in which trajectory planning itself is formulated directly as an MPC optimization problem. Such formulations can explicitly include state constraints, actuator limitations, obstacle constraints, and dynamic feasibility within the trajectory-generation process [165]. The resulting trajectories are inherently compatible with the vehicle's motion dynamics, reducing the mismatch between planning and control.

7.5 Cooperative Unmanned Surface Vehicle–Unmanned Underwater Vehicle Systems

Early research on USV–UUV systems primarily focused on system-level integration and communication. Integrated surface–underwater platforms have been developed in which the USV serves as a surface hub for communication, power supply, localization, and data collection while the underwater vehicle performs subsea sensing [166]. These systems demonstrated the practical value of heterogeneous cooperation but also revealed the challenges associated with relative localization, communication, and coupled motion.

Subsequent studies investigated cooperative trajectory tracking using leader–follower and consensus-based methods. Jia et al. [167] proposed a robust distributed cooperative rendezvous controller for heterogeneous marine vehicles based on MPC. The approach considers input limitations, inter-vehicle safety constraints, external disturbances, and networked communication while allowing individual vehicles to solve local optimization problems. This study represents an important step toward distributed MPC for heterogeneous surface–underwater systems.

Another emerging direction involves tethered USV–UUV/ROV cooperation. In such systems, the USV and underwater vehicle are physically coupled through a tether, introducing additional geometric and dynamic constraints. A multi-objective NMPC framework was recently proposed for tethered USV–ROV cooperative tracking and dynamic obstacle avoidance [168]. The controller simultaneously considered the predicted three-dimensional trajectory of the underwater vehicle, USV nonholonomic dynamics, surface obstacles, tether-length constraints, and control

smoothness. This type of formulation demonstrates the ability of MPC to address strongly coupled heterogeneous systems within a unified predictive framework.

Recent research has also considered surface–underwater joint observation. A real-time optimization-enhanced MPC framework was developed for a joint USV–underwater observation system, incorporating adaptive optimization and fuzzy parameter adjustment to improve trajectory tracking under complex marine conditions [169]. Such work illustrates an emerging transition from independent vehicle control toward mission-oriented coordination of heterogeneous sensing platforms [170, 171].

8 Key Challenges and Future Directions

8.1 Modeling Uncertainty

The performance of MPC depends strongly on the accuracy of its prediction model. Transportation systems, however, are nonlinear, time-varying, and exposed to operating conditions that cannot always be described accurately in advance. In autonomous road vehicles, tire–road friction, vehicle parameters, and the behavior of surrounding road users may change during operation. UAV dynamics are affected by aerodynamic effects, payload variation, and wind disturbances, while marine vehicles are subject to uncertain hydrodynamic coefficients, waves, currents, and wind. At the traffic-network level, travel demand, turning ratios, queue propagation, and route-choice behavior are also difficult to predict reliably.

Model mismatch can produce inaccurate state predictions, reduce tracking or coordination performance, and lead to constraint violations. A central research problem is therefore how to update the prediction model without losing the physical structure required for analysis and verification. Physics-based models can be combined with online parameter estimation, system identification, and data-driven residual correction. In such formulations, the physical model describes the principal system dynamics, while the learned component represents effects that are difficult to model analytically. Future studies should examine not only prediction accuracy but also identifiability, uncertainty bounds, data requirements, recursive feasibility, and closed-loop stability. These issues are especially important when an adaptive or learned model is used in safety-critical transportation operations.

8.2 Computational Complexity

Online optimization remains a major obstacle to the real-time implementation of MPC in intelligent transportation systems. Computational demand generally increases with the prediction horizon, model dimension, degree of nonlinearity, number of interacting agents, and complexity of safety and operational constraints. Mixed-integer decisions, nonconvex collision-avoidance constraints, and distributed coordination can further increase solution time. This problem is particularly important for high-speed road vehicles and UAVs, where control commands may need to be updated within a few tens of milliseconds, but it also arises in large railway and traffic networks because of their spatial scale and number of decision variables.

Model reduction, successive linearization, explicit MPC, warm starts, real-time iteration schemes, parallel computing, distributed optimization, and event-triggered updates have been used to reduce online computation. Learning-based approximations of optimization policies have also received increasing attention, although their behavior outside the training distribution remains difficult to verify. Further work is needed on solver–controller co-design, hardware-aware optimization, adaptive horizon selection, decomposition of large-scale problems, and certified approximate solutions. Computational performance should be reported using reproducible measures such as worst-case solution time, deadline-miss rate, memory consumption, processor type, and scaling with the number of agents. Average computation time alone is insufficient for assessing whether an MPC scheme can operate reliably under strict real-time requirements.

8.3 Multi-Objective Optimization

Transportation control generally involves several competing objectives, including safety, mobility, energy use, passenger comfort, punctuality, emissions, and infrastructure utilization. Faster acceleration may shorten travel time but increase energy consumption and passenger discomfort. Similarly, conservative collision-avoidance behavior may increase separation margins while reducing traffic throughput or causing unnecessary deviations from the planned route. The relative importance of these objectives also changes with traffic density, environmental conditions, mission stage, and operating risk.

Most existing MPC formulations combine different objectives through fixed weighting coefficients. Although this approach is convenient, the selected weights may not remain appropriate across all operating conditions, and their physical interpretation is often unclear. Future research should examine Pareto-based formulations, lexicographic optimization, hierarchical objectives, risk-sensitive costs, and context-dependent weight adaptation. Particular attention should be given to preventing adaptive weights from weakening safety-related constraints. It is also necessary to distinguish non-negotiable safety requirements from performance objectives that may be traded against one another. This distinction would make multi-objective MPC more transparent and suitable for system-level

transportation management, where the interests of individual vehicles, passengers, infrastructure operators, and the wider network may not coincide.

8.4 Complex Constraint Coupling

Intelligent transportation systems are governed by physical, operational, communication, and safety constraints that are often strongly coupled. In autonomous vehicles, tire–road friction links longitudinal acceleration, braking, and lateral motion. In railway systems, headway constraints interact with braking distance, speed profiles, station dwell times, and timetable recovery. For UAVs, actuator limits affect trajectory tracking, collision avoidance, and formation maintenance simultaneously. In marine navigation, vessel dynamics, encounter geometry, obstacle boundaries, and COLREGs-related maneuvering requirements must be considered together.

MPC can represent these constraints within a single optimization problem, but tightly coupled formulations may become nonconvex, computationally expensive, or infeasible under unexpected operating conditions. Future work should investigate tractable constraint reformulations, decomposition methods, constraint prioritization, adaptive safety margins, and feasibility-restoration mechanisms. Formal verification and runtime safety supervision are also needed when an MPC controller is combined with learned prediction models or approximate solvers. At the system level, research should account for constraints arising from infrastructure capacity, communication availability, mixed human–autonomous traffic, and interactions among independently controlled agents. Reliable deployment will depend on whether these coupled constraints can be handled within the available computation time while maintaining clearly stated feasibility and safety conditions.

8.5 Simulation-to-Reality Gap

A substantial part of the MPC literature in intelligent transportation is still evaluated primarily through numerical simulation. Simulation permits controlled and repeatable comparisons, but it cannot fully reproduce sensor errors, actuator dynamics, communication delays, packet loss, processor limitations, environmental disturbances, model aging, and unexpected behavior by human operators or other traffic participants. As a result, performance obtained under idealized simulation conditions may not be maintained on physical platforms or in operational transportation networks.

Validation should therefore progress through several levels, including numerical simulation, software-in-the-loop testing, processor-in-the-loop testing, hardware-in-the-loop experiments, controlled physical trials, and field deployment. At each level, studies should report not only tracking accuracy or objective-function values but also computation time, constraint violations, failure cases, recovery behavior, communication demand, and sensitivity to model and sensor errors. Benchmark scenarios and shared datasets would also support more reliable comparisons among MPC variants. This need is particularly evident in UAV swarms and marine systems, where communication interruptions and environmental disturbances are common, but it applies equally to connected road vehicles, railway operations, and network-level traffic control. Closing the simulation-to-reality gap requires closer integration among controller design, sensing, communication, embedded implementation, infrastructure operation, and safety assessment.

9 Conclusions

This paper reviewed the development and application of MPC across ground, railway, low-altitude aerial, and marine unmanned transportation systems. The cross-domain analysis showed that MPC provides a common framework for predictive decision-making, receding-horizon feedback, multi-objective optimization, and explicit constraint handling, although its formulation and implementation requirements differ substantially among transportation modes. Ground-vehicle applications generally require rapid responses to dense interactions and changing road conditions, whereas railway applications place greater emphasis on energy-efficient operation, timetable coordination, headway management, and network-scale scheduling. UAV applications are dominated by nonlinear flight dynamics, limited onboard computation, and three-dimensional multi-agent coordination. Marine applications must additionally address underactuation, uncertain hydrodynamics, environmental disturbances, navigation rules, and restricted surface–underwater communication.

Across these domains, MPC has been studied for trajectory tracking, motion regulation, collision avoidance, energy management, traffic control, scheduling, and cooperative operation. The reviewed literature also indicates a gradual shift from isolated platform control toward integrated planning and control, distributed coordination, and transportation-system-level optimization. Nevertheless, the maturity of these applications varies considerably. Many proposed methods have been demonstrated only in simulation or controlled experiments, and direct comparison remains difficult because studies use different models, operating scenarios, performance measures, computing platforms, and assumptions about uncertainty.

Several common barriers continue to limit broader deployment, including model uncertainty, online computational demand, competing operational objectives, tightly coupled constraints, communication limitations,

and the gap between simulation and field performance. Addressing these barriers requires more than further development of optimization algorithms. Future research should connect MPC design with sensing and state estimation, communication architecture, embedded computing, infrastructure management, formal safety assessment, and staged experimental validation. Physics-informed learning, robust and stochastic formulations, distributed and hierarchical MPC, and hardware-aware optimization are promising directions, but their practical value must be established through transparent assumptions, reproducible benchmarks, and realistic operating tests.

Overall, MPC offers a coherent basis for coordinating prediction, control, and operational decision-making across heterogeneous transportation modes. Its future role in intelligent transportation will depend on whether theoretical guarantees, computational feasibility, and field-level reliability can be achieved simultaneously. Closer integration of vehicle-level control with traffic management, infrastructure coordination, and cooperative system operation will be central to translating MPC research into deployable intelligent transportation systems.

Author Contributions

Conceptualization, Y.G.S. and Y.J.C.; methodology, B.R., H.T.X., and D.D.Z.; investigation, B.R., H.T.X., and D.D.Z.; writing—original draft preparation, B.R., H.T.X., and D.D.Z.; writing—review and editing, Y.G.S. and Y.J.C.; supervision, Y.G.S. and Y.J.C.; project administration, Y.G.S. and Y.J.C. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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